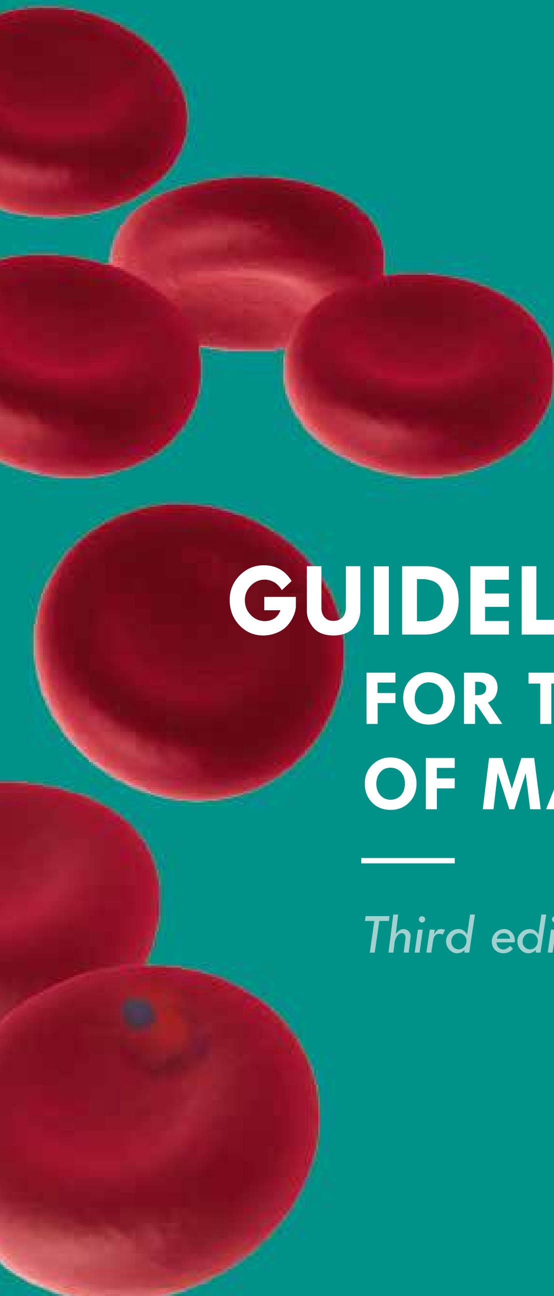

GUIDELINES FOR THE TREATMENT OF MALARIA

Third edition

World Health Organization

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Contents

Glossary

Artemisinin-based combination therapy (ACT). A combination of an artemisinin derivative with a longer-acting antimalarial that has a different mode of action.

Asexual cycle. The life cycle of the malaria parasite in the host, from merozoite invasion of red blood cells to schizont rupture (merozoite → ring stage → trophozoite → schizont → merozoites). The duration is approximately 24 h in *Plasmodium knowlesi*, 48 h in *P. falciparum*, *P. ovale* and *P. vivax* and 72 h in *P. malariae*.

Asexual parasitaemia. The presence of asexual parasites in host red blood cells. The level of asexual parasitaemia determined by microscopy can be expressed in several ways: the percentage of infected red blood cells, the number of infected red cells per unit volume of blood, the number of parasites seen in one field on high power microscopy examination of a thick blood film, or the number of parasites seen per 200–1000 white blood cells on high-power examination of a thick blood film.

Asymptomatic parasitaemia. The presence of asexual parasites in the blood without symptoms of illness.

Cerebral malaria. Severe *P. falciparum* malaria with coma (Glasgow coma scale < 11, Blantyre coma scale < 3); malaria with coma persisting for > 30 min after a seizure.

Combination treatment. A combination of two or more classes of antimalarial drug with unrelated mechanisms of action.

Cure. Elimination of the malaria parasites that caused the treated illness.

Drug resistance. The ability of a parasite strain to survive and/or to multiply despite the administration and absorption of a drug given in doses equal to or higher than those usually recommended, provided the exposure is adequate. Resistance to antimalarial agents arises because of the selection of parasites with genetic changes (mutations or gene amplifications) that confer reduced susceptibility.

Gametocytes. The sexual stages of malaria parasites that infect anopheline mosquitoes when taken up during a blood meal.

Fixed-dose combination. A combination in which two antimalarial drugs are formulated together in the same tablet, capsule, powder, suspension or granule.

Hyperparasitaemia. A high density of parasites in the blood, which increases the risk of deterioration to severe malaria (although the risk varies in different endemic areas according to the level of transmission) and of subsequent treatment failure. In this document, the term is used to refer to a parasite density > 4% (~ 200 000/µL). Patients with *P. falciparum* parasite densities > 10% and patients with *P. knowlesi* parasite densities > 100 000/µL (~ 2%) are considered to have severe malaria even if they do not have evidence of vital organ dysfunction.

Hypnozoites. Persistent liver stages of *P. vivax* and *P. ovale* malaria that remain dormant in host hepatocytes for 3–45 weeks before maturing to form hepatic schizonts, which then burst and release merozoites that infect red blood cells. This is the cause of relapses.

Malaria pigment (haemozoin). A dark-brown granular material formed by malaria parasites as a by-product of haemoglobin digestion. Pigment is evident in mature trophozoites and schizonts. It may also be phagocytosed by monocytes, macrophages and polymorphonuclear neutrophils.

Merozoite. Parasite released into the host bloodstream when a hepatic or erythrocytic schizont bursts. The merozoites then invade red blood cells.

Monotherapy. Antimalarial treatment with a single medicine: either a single active compound or a synergistic combination of two compounds with related mechanisms of action.

***Plasmodium*.** A genus of protozoan vertebrate blood parasites that includes the causal agents of malaria. *P. falciparum*, *P. malariae*, *P. ovale* (two species) and *P. vivax* cause malaria in humans. Human infections with the monkey malaria parasite *P. knowlesi* and very occasionally with other simian malaria species may occur in forested regions of South-East Asia.

Pre-erythrocytic development. The development of the malaria parasite when it first enters the host. After inoculation into a human by a female anopheline mosquito, sporozoites invade hepatocytes in the host liver and multiply there for 5–12 days, forming hepatic schizonts. These then burst, liberating merozoites into the bloodstream, where they subsequently invade red blood cells.

Radical cure. This term refers to both cure of blood-stage infection and prevention of relapses by killing hypnozoites (in *P. vivax* and *P. ovale* infections only).

Rapid diagnostic test (RDT). An antigen-based stick, cassette or card test for malaria in which a coloured line indicates the presence of plasmodial antigens.

Recrudescence. Recurrence of asexual parasitaemia following antimalarial treatment comprising the same genotype(s) that caused the original illness. This results from incomplete clearance of asexual parasitaemia because of inadequate or ineffective treatment. It must be distinguished from re-infection (usually determined by molecular genotyping in endemic areas). Recrudescence is different from relapse in *P. vivax* and *P. ovale* infections.

Recurrence. Recurrence of asexual parasitaemia after treatment, due to recrudescence, relapse (in *P. vivax* and *P. ovale* infections only) or a new infection.

Relapse. Recurrence of asexual parasitaemia in *P. vivax* and *P. ovale* malaria deriving from persisting liver stages; occurs when the blood-stage infection has been eliminated but hypnozoites persist in the liver and mature to form hepatic schizonts. After an interval of weeks or months, the hepatic schizonts burst and liberate merozoites into the bloodstream.

Ring stage. Young, usually ring-shaped, intra-erythrocytic malaria parasites, before malaria pigment is evident by microscopy.

Schizont. Mature malaria parasite in host liver cells (hepatic schizont) or red blood cells (erythrocytic schizont) that is undergoing nuclear division by a process called schizogony.

Selection pressure. Resistance to antimalarial agents emerges and spreads because of the survival advantage of resistant parasites in the presence of the drug. The selection pressure reflects the intensity and magnitude of selection: the greater the proportion of parasites in a given population exposed to concentrations of an antimalarial agent that allow proliferation of resistant but not sensitive parasites, the greater is the selection pressure.

Severe anaemia. Haemoglobin concentration of < 5 g/100 mL (haematocrit < 15%).

Severe falciparum malaria. Acute falciparum malaria with signs of severity and/or evidence of vital organ dysfunction.

Sporozoite. Motile malaria parasite that is infective to humans, inoculated by a feeding female anopheline mosquito, that invades hepatocytes.

Transmission intensity. This is the frequency with which people living in an area are bitten by anopheline mosquitoes carrying human malaria sporozoites. It is often expressed as the annual entomological inoculation rate, which is the average number of inoculations with malaria parasites received by one person in 1 year.

Trophozoite. The stage of development of malaria parasites growing within host red blood cells from the ring stage to just before nuclear division. Mature trophozoites contain visible malaria pigment.

Uncomplicated malaria. Symptomatic malaria parasitaemia with no signs of severity and/or evidence of vital organ dysfunction.

Vectorial capacity. Number of potential new infections that the population of a given anopheline mosquito vector would distribute per malaria case per day at a given place and time.

Abbreviations

ACT	artemisinin-based combination therapy
bw	body weight
CI	confidence interval
DTP	diphtheria, tetanus and pertussis (vaccine)
GRADE	Grading of Recommendations Assessment, Development and Evaluation
G6PD	glucose-6-phosphate dehydrogenase
HRP2	histidine-rich protein 2
IPTp	intermittent preventive treatment in pregnancy
IPTi	intermittent preventive treatment in infancy
PCR	polymerase chain reaction
*Pf*HRP2	*Plasmodium falciparum* histidine-rich protein-2
*p*LDH	*parasite*-lactate dehydrogenase
Pvdhfr	*Plasmodium vivax* dihydrofolate reductase gene
RDT	rapid diagnostic test
RR	relative risk, or risk ratio
SMC	seasonal malaria chemoprevention
SP	sulfadoxine–pyrimethamine

3RD

EDITION

EXECUTIVE SUMMARY

Malaria case management, consisting of early diagnosis and prompt effective treatment, remains a vital component of malaria control and elimination strategies. This third edition of the WHO *Guidelines for the treatment of malaria* contains updated recommendations based on new evidence particularly related to dosing in children, and also includes recommendations on the use of drugs to prevent malaria in groups at high risk.

Core principles

The following core principles were used by *the Guidelines* Development Group that drew up these *Guidelines*.

1. Early diagnosis and prompt, effective treatment of malaria

Uncomplicated falciparum malaria can progress rapidly to severe forms of the disease, especially in people with no or low immunity, and severe falciparum malaria is almost always fatal without treatment. Therefore, programmes should ensure access to early diagnosis and prompt, effective treatment within 24–48 h of the onset of malaria symptoms.

2. Rational use of antimalarial agents

To reduce the spread of drug resistance, limit unnecessary use of antimalarial drugs and better identify other febrile illnesses in the context of changing malaria epidemiology, antimalarial medicines should be administered only to patients who truly have malaria. Adherence to a full treatment course must be promoted. Universal access to parasitological diagnosis of malaria is now possible with the use of quality-assured rapid diagnostic tests (RDTs), which are also appropriate for use in primary health care and community settings.

3. Combination therapy

Preventing or delaying resistance is essential for the success of both national and global strategies for control and eventual elimination of malaria. To help protect current and future antimalarial medicines, all episodes of malaria should be treated with at least two effective antimalarial medicines with different mechanisms of action (combination therapy).

4. Appropriate weight-based dosing

To prolong their useful therapeutic life and ensure that all patients have an equal chance of being cured, the quality of antimalarial drugs must be ensured and antimalarial drugs must be given at optimal dosages. Treatment should maximize the likelihood of rapid clinical and parasitological cure and minimize transmission from the treated infection. To achieve this, dosage regimens should be based on the patient's weight and should provide effective concentrations of antimalarial drugs for a sufficient time to eliminate the infection in all target populations.

Recommendations

Diagnosis of malaria

All cases of suspected malaria should have a parasitological test (microscopy or Rapid diagnostic test (RDT)) to confirm the diagnosis.

Both microscopy and RDTs should be supported by a quality assurance programme.
Good practice statement

Treating uncomplicated *P. falciparum* malaria

Treatment of uncomplicated P. falciparum malaria

Treat children and adults with uncomplicated *P. falciparum* malaria (except pregnant women in their first trimester) with one of the following recommended artemisinin-based combination therapies (ACT):

- artemether + lumefantrine
- artesunate + amodiaquine
- artesunate + mefloquine
- dihydroartemisinin + piperaquine
- artesunate + sulfadoxine–pyrimethamine (SP)

Strong recommendation, high-quality evidence

Duration of ACT treatment

ACT regimens should provide 3 days' treatment with an artemisinin derivative.
Strong recommendation, high-quality evidence

Revised dose recommendation for dihydroartemisinin + piperaquine in young children

Children < 25kg treated with dihydroartemisinin + piperaquine should receive a minimum of 2.5 mg/kg body weight (bw) per day of dihydroartemisinin and 20 mg/kg bw per day of piperaquine daily for 3 days.
Strong recommendation based on pharmacokinetic modelling

Reducing the transmissibility of treated P. falciparum infections

In low-transmission areas, give a single dose of 0.25 mg/kg bw primaquine with ACT to patients with *P. falciparum* malaria (except pregnant women, infants aged < 6 months and women breastfeeding infants aged < 6 months) to reduce transmission. Testing for glucose-6-phosphate dehydrogenase (G6PD) deficiency is not required.
Strong recommendation, low-quality evidence

Treating uncomplicated *P. falciparum* malaria in special risk groups

First trimester of pregnancy

Treat pregnant women with uncomplicated *P. falciparum* malaria during the first trimester with 7 days of quinine + clindamycin.
Strong recommendation

Infants less than 5kg body weight

Treat infants weighing < 5 kg with uncomplicated *P. falciparum* malaria with ACT at the same mg/kg bw target dose as for children weighing 5 kg.
Strong recommendation

Patients co-infected with HIV

In people who have HIV/AIDS and uncomplicated *P. falciparum* malaria, avoid artesunate + SP if they are being treated with co-trimoxazole, and avoid artesunate + amodiaquine if they are being treated with efavirenz or zidovudine.
Good practice statement

Non-immune travellers

Treat travellers with uncomplicated *P. falciparum* malaria returning to non-endemic settings with ACT.
Strong recommendation, high-quality evidence

Hyperparasitaemia

People with *P. falciparum* hyperparasitaemia are at increased risk for treatment failure, severe malaria and death and should be closely monitored, in addition to receiving ACT.
Good practice statement

Treating uncomplicated *P. vivax*, *P. ovale*, *P. malariae* or *P. knowlesi* malaria

Blood stage infection

If the malaria species is not known with certainty, treat as for uncomplicated *P. falciparum* malaria.
Good practice statement

In areas with chloroquine-susceptible infections, treat adults and children with uncomplicated *P. vivax*, *P. ovale*, *P. malariae* or *P. knowlesi* malaria with either ACT (except pregnant women in their first trimester) or chloroquine.
Strong recommendation, high-quality evidence

In areas with chloroquine-resistant infections, treat adults and children with uncomplicated *P. vivax*, *P. ovale*, *P. malariae* or *P. knowlesi* malaria (except pregnant women in their first trimester) with ACT.
Strong recommendation, high-quality evidence

Treat pregnant women in their first trimester who have chloroquine-resistant *P. vivax* malaria with quinine.
Strong recommendation, very low-quality evidence

Preventing relapse in *P. vivax* or *P. ovale* malaria

The G6PD status of patients should be used to guide administration of primaquine for preventing relapse.
Good practice statement

To prevent relapse, treat *P. vivax* or *P. ovale* malaria in children and adults (except pregnant women, infants aged < 6 months, women breastfeeding infants aged < 6 months, women breastfeeding older infants unless they are known not to be G6PD deficient, and people with G6PD deficiency) with a 14-day course (0.25-0.5 mg/kg bw daily) of primaquine in all transmission settings.
Strong recommendation, high-quality evidence

In people with G6PD deficiency, consider preventing relapse by giving primaquine base at 0.75 mg/kg bw once a week for 8 weeks, with close medical supervision for potential primaquine-induced haemolysis.
Conditional recommendation, very low-quality evidence

When G6PD status is unknown and G6PD testing is not available, a decision to prescribe primaquine must be based on an assessment of the risks and benefits of adding primaquine.
Good practice statement

Pregnant and breastfeeding women

In women who are pregnant or breastfeeding, consider weekly chemoprophylaxis with chloroquine until delivery and breastfeeding are completed, then, on the basis of G6PD status, treat with primaquine to prevent future relapse.
Conditional recommendation, moderate-quality evidence

Treating severe malaria

Treat adults and children with severe malaria (including infants, pregnant women in all trimesters and lactating women) with intravenous or intramuscular artesunate for at least 24 h and until they can tolerate oral medication. Once a patient has received at least 24 h of parenteral therapy and can tolerate oral therapy, complete treatment with 3 days of ACT (add single dose primaquine in areas of low transmission).
Strong recommendation, high-quality evidence

Revised dose recommendation for parenteral artesunate in young children

Children weighing < 20 kg should receive a higher dose of artesunate (3 mg/kg bw per dose) than larger children and adults (2.4 mg/kg bw per dose) to ensure equivalent exposure to the drug.
Strong recommendation based on pharmacokinetic modelling

Parenteral alternatives where artesunate is not available

If artesunate is not available, use artemether in preference to quinine for treating children and adults with severe malaria.
Conditional recommendation, low-quality evidence

Treating cases of suspected severe malaria pending transfer to a higher-level facility (pre-referral treatment)

Pre-referral treatment options

Where complete treatment of severe malaria is not possible but injections are available, give adults and children a single intramuscular dose of artesunate, and refer to an appropriate facility for further care. Where intramuscular artesunate is not available use intramuscular artemether or, if that is not available, use intramuscular quinine.
Strong recommendation, moderate-quality evidence

Where intramuscular injection of artesunate is not available, treat children < 6 years with a single rectal dose (10mg/kg bw) of artesunate, and refer immediately to an appropriate facility for further care. Do not use rectal artesunate in older children and adults.
Strong recommendation, moderate-quality evidence

Chemoprevention for special risk groups

Intermittent preventive treatment in pregnancy

In malaria-endemic areas in Africa, provide intermittent preventive treatment with SP to all women in their first or second pregnancy (SP-IPTp) as part of antenatal care. Dosing should start in the second trimester and doses should be given at least 1 month apart, with the objective of ensuring that at least three doses are received.
Strong recommendation, high-quality evidence

Intermittent preventive treatment in infants

In areas of moderate-to-high malaria transmission of Africa, where SP is still effective, provide intermittent preventive treatment with SP to infants (< 12 months of age) (SP-IPTi) at the time of the second and third rounds of vaccination against diphtheria, tetanus and pertussis (DTP) and vaccination against measles.
Strong recommendation

Seasonal malaria chemoprevention

In areas with highly seasonal malaria transmission in the sub-Sahel region of Africa, provide seasonal malaria chemoprevention (SMC) with monthly amodiaquine + SP for all children aged < 6 years during each transmission season.
Strong recommendation, high-quality evidence

Antimalarial drug quality

National drug and regulatory authorities should ensure that the antimalarial medicines provided in both the public and the private sectors are of acceptable quality, through regulation, inspection and law enforcement.
Good practice statement

Monitoring the efficacy of antimalarial drugs

All malaria programmes should regularly monitor the therapeutic efficacy of antimalarial drugs using the standard WHO protocols.
Good practice statement

National adaptation and implementation

The choice of ACTs in a country or region should be based on optimal efficacy, safety and adherence.
Good practice statement

Drugs used in IPTp, SMC and IPTi should not be used as a component of first-line treatments in the same country or region.
Good practice statement

When possible, use:

- fixed-dose combinations rather than co-blistered or loose, single-agent formulations; and
- for young children and infants, paediatric formulations, with a preference for solid formulations (e.g. dispersible tablets) rather than liquid formulations.

Good practice statement

3RD EDITION

1 | INTRODUCTION

1.1 | BACKGROUND

Malaria remains an important cause of illness and death in children and adults in countries in which it is endemic. Malaria control requires an integrated approach, including prevention (primarily vector control) and prompt treatment with effective antimalarial agents. Since publication of the first edition of *the Guidelines for the treatment of malaria* in 2006 and the second edition in 2010, all countries in which *P. falciparum* malaria is endemic have progressively updated their treatment policy from use of monotherapy with drugs such as chloroquine, amodiaquine and sulfadoxine–pyrimethamine (SP) to the currently recommended artemisinin-based combination therapies (ACT). The ACTs are generally highly effective and well tolerated. This has contributed substantially to reductions in global morbidity and mortality from malaria. Unfortunately, resistance to artemisinins has arisen recently in *P. falciparum* in South-East Asia, which threatens these gains.

The treatment recommendations in this edition of *the Guidelines* have a firm evidence base for most antimalarial drugs, but, inevitably, there are still information gaps. *The Guidelines* will therefore remain under regular review, with updates every 2 years or more frequently as new evidence becomes available. The treatment recommendations in the main document are brief; for those who wish to study the evidence base in more detail, a series of annexes is provided, with references to the appropriate sections of the main document.

1.2 | OBJECTIVES

The objectives of *the Guidelines* are to:

- assist policy-makers to design and refine effective national treatment policies on the basis of the best available evidence;
- help hospital and clinical care providers to design and refine effective treatment protocols on the basis of the best available evidence;
- promote the use of safe, effective malaria treatment; and
- protect currently effective malaria treatment against the development of resistance.

1.3 | SCOPE

The Guidelines provide a framework for designing specific, detailed national treatment protocols, taking into account local patterns of resistance to antimalarial drugs and health service capacity.

The Guidelines provide evidence-based recommendations on:

- the treatment of uncomplicated and severe malaria in all age groups and situations, including in young children, pregnant women, people who are HIV positive, travellers from non-malaria-endemic regions and in epidemics and complex emergency situations; and
- the use of antimalarial drugs as preventive therapy in healthy people living in malaria-endemic areas who are high risk, in order to reduce morbidity and

mortality from malaria. The two approaches used are intermittent preventive treatment (IPT) and seasonal malaria chemoprevention (SMC).

No guidance is given in this edition on the use of antimalarial agents to prevent malaria in people travelling from non-endemic settings to areas of malaria transmission. This is available in the WHO, International travel and health guidance.[1]

1.4 | TARGET AUDIENCE

The Guidelines are designed primarily for policy-makers in ministries of health, who formulate country-specific treatment guidelines. Other groups that may find them useful include health professionals (doctors, nurses and paramedical officers) and public health and policy specialists working in hospitals, research institutions, medical schools, non-governmental organizations and agencies that are partners in health or malaria control, the pharmaceutical industry and primary health-care services.

1.5 | METHODS USED TO MAKE THE RECOMMENDATIONS

This third edition of *the Guidelines for the treatment of malaria* was prepared in accordance with the latest WHO standard methods for guideline development.[2] This involves planning, "scoping" and needs assessment, establishment of a guideline development group, formulation of key questions (population, participants or patients; intervention or indicator; comparator or control; outcome: "PICO"), commissioning reviews, applying Grading of Recommendations Assessment, Development and Evaluation (GRADE) to the quality of the evidence and making recommendations.[3] This method (see Annex 1) ensures a transparent link between the evidence and the recommendations.

1.5.1 | SOURCES OF EVIDENCE

After the scoping meeting, the Cochrane Infectious Diseases Group at the Liverpool School of Tropical Medicine in Liverpool, England, was commissioned to undertake systematic reviews and to assess the quality of the evidence for each priority question. All the reviews involved extensive searches for published and unpublished trials and highly sensitive searches of the Cochrane Infectious Diseases Group trials register, the Cochrane Central Register of Controlled Trials, MEDLINE®, Embase and LILACS. All the reviews are published on line in the Cochrane Library.

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1 WHO International travel and health. Geneva : World Health Organization; 2014 (http://www.who.int/ith/en/).

2 WHO handbook for guideline development. Geneva: World Health Organization; 2012 (http://www.who.int/kms/guidelines_review_committee/en/).

3 Guyatt GH, Oxman AD, Vist G, Kunz R, Falck-Ytter Y, Alonso-Coello P, Schünemann HJ, for the GRADE Working Group. Rating quality of evidence and strength of recommendations GRADE: an emerging consensus on rating quality of evidence and strength of recommendations 2008. BMJ 336:924-926.

When little evidence was available from randomized trials, the group considered published reviews of non-randomized studies.

A subgroup on dosing reviewed published studies from MEDLINE® and Embase on the pharmacokinetics and pharmacodynamics of antimalarial medicines. They also used raw data from the WorldWide Antimalarial Resistance Network, a repository of clinical and laboratory data on pharmacokinetics and dosing simulations in individual patients, including measurements using validated assays of concentrations of antimalarial medicines in plasma or whole blood. The data came either from peer-reviewed publications or were submitted to regulatory authorities for drug registration. Population pharmacokinetics models were constructed, and the concentration profiles of antimalarial medicines in plasma or whole blood were simulated (typically 1000 times) for each weight category to inform dose recommendations.

1.5.2 | QUALITY OF EVIDENCE

The quality of the evidence from systematic reviews was assessed for each outcome and rated on a four-point scale, after consideration of the risk for bias (including publication bias) and the consistency, directness and precision of the effect estimates. The terms used in the quality assessments refer to the confidence that the guideline development group had in the estimate and not to the scientific quality of the investigations reviewed:

Quality of evidence	Interpretation
High	The group is very confident in the estimates of effect and considers that further research is very unlikely to change this confidence.
Moderate	The group has moderate confidence in the estimate of effect but considers that further research is likely to have an important impact on their confidence and may change the estimate.
Low	The group has low confidence in the estimate of effect and considers that further research is very likely to have an important impact on their confidence and is likely to change the estimate.
Very low	The group is very uncertain about the estimate of effect.

1.5.3 | FORMULATION OF RECOMMENDATIONS

The systematic reviews, the GRADE and other relevant materials were securely provided to all members of the Guideline Development Group. Recommendations were formulated after considering the quality of the evidence, the balance of benefits and harm and the feasibility of the intervention based on the four core principles listed in the executive summary. Although cost is a critical factor in setting national antimalarial treatment policies, cost was not formally considered. The dose recommendations were designed to ensure equivalent exposure of all patient groups to the drug. A revised dose regimen was recommended when there was sufficient evidence that the dose should be changed in order to achieve the target exposure. The Guideline Development Group discussed both the proposed wording of the recommendations and the rating of its strength. Areas of disagreement were resolved through extensive discussions at the meetings, e-mail and teleconferencing. The final draft was circulated to the Guideline Development Group and external peer reviewers. The external comments were addressed where possible and incorporated into the revised guidelines. Consensus was reached on all the recommendations, strength of evidence and the wording of *the guidelines*. Voting was not required at any stage.

Factor considered	Rationale
Balance of benefits and harm	The more the expected benefits outweigh the expected risks, the more likely it is that a strong recommendation will be made. When the balance of benefits and harm is likely to vary by setting or is a fine balance, a conditional recommendation is more likely.
Values and preferences	If the recommendation is likely to be widely accepted or highly valued, a strong recommendation is more likely.
Feasibility	If an intervention is achievable in the settings in which the greatest impact is expected, a strong recommendation is more likely.

1.5.4 | TYPES OF RECOMMENDATIONS

Three types of recommendations are presented in *the guidelines*

- *Treatment recommendations:* These recommendations were formulated using the GRADE approach, supported by systematic reviews of the evidence, with formal assessment of the quality of evidence, that were used by the panel in making the recommendations. However, a small number of treatment recommendations are not accompanied by GRADE tables, and these have been labelled as 'evidence not graded'. These recommendations were made when the panel considered there to be such limited evidence available on alternatives to current practice that they could do little but recommend the status quo pending further research.

- *Dosing recommendations:* These recommendations were formulated using mathematical modelling, based on summaries of systematically collected pharmacokinetic data, to predict drug exposures in people of different body-weights, particularly those who are generally under-represented in clinical trials such as young infants.
- *Good practice statements:* These statements reflect a consensus among the panel, that the net benefits of adherence to the statement are large, unequivocal and the implications of the statement are common sense. These statements are made to re-emphasize the basic principles of good care, or good management practice with implementation, such as quality assurance of antimalarial medicines.

1.5.5 | STRENGTH OF RECOMMENDATIONS

Each recommendation was then classified as strong or conditional:

Strength of recommendation	Interpretation		
	For policy-makers	For clinicians	For patients
Strong	This recommendation can be adopted as policy in most situations.	Most individuals should receive the recommended course of action.	Most people in your situation would want the recommended course of action.
Conditional	Substantial debate should be conducted at national level, with the involvement of various stakeholders.	Be prepared to help individuals in making a decision that is consistent with their own values.	The majority of people in your situation would want the recommended course of action, but many would not.

1.5.6 | PRESENTATION OF EVIDENCE (RECOMMENDATIONS)

For clarity, *the guidelines* are presented in a simple descriptive form in the main document, with the recommendations. The recommendations are summarized in boxes at the start of each section (green), an evidence box (blue) is presented for each of the recommendation with a link made to the the GRADE profiles and decision tables. The complete GRADE tables, the pharmacokinetics underlying dose revisions, and additional references are provided in annexe 4.

1.5.7 | FUNDING

The preparation and printing of *the guidelines* were funded exclusively by the WHO Global Malaria Programme. No external source of funding either from bilateral technical partners or from industry was solicited or used.

1.5.8 | MANAGEMENT OF CONFLICTS OF INTEREST

All members of the guideline development group and external expert reviewers made declarations of interest, which were managed in accordance with WHO procedures and cleared by the Legal Department. The WHO Guideline Steering Group and the co-chairs of *the Guidelines* Development Group were satisfied that there had been a transparent declaration of interests. No case necessitated the exclusion of any of the Guideline Development Group member or an external peer reviewer. The members of the guideline development group and a summary of declaration of interest listed in Annex 1.

1.5.9 | DISSEMINATION

The guidelines will be disseminated as a printed publication and electronically on the WHO web site in three languages (English, French and Spanish). A library of all supporting documentation will be also available on the web site. WHO headquarters will work closely with the regional and country offices to ensure the wide dissemination of *the guidelines* to all malaria endemic countries. There will also be dissemination through regional, sub-regional and country meetings. Member States will be supported to adapt and implement these guidelines (further details on national adaptation and implementation provided in Chapter 14).

1.5.10 | UPDATING

The evidence will be reviewed regularly and updated every 2 years or more frequently as new evidence becomes available. A mechanism will be established for periodic monitoring and evaluation of use of the treatment guidelines in countries.

1.5.11 | USER FEEDBACK

An online survey was carried out to obtain feedback from users of the second edition of *the Guidelines for the treatment of malaria*, and the responses were used in updating this edition.

3RD EDITION

2 | CLINICAL MALARIA AND EPIDEMIOLOGY

2.1 | ETIOLOGY AND SYMPTOMS

Malaria is caused by infection of red blood cells with protozoan parasites of the genus *Plasmodium* inoculated into the human host by a feeding female anopheline mosquito. The five human *Plasmodium* species transmitted from person to person are *P. falciparum*, *P. vivax*, *P. ovale* (two species) and *P. malariae*. Increasingly, human infections with the monkey malaria parasite *P. knowlesi* are being reported from the forested regions of South-East Asia and particularly the island of Borneo.

The first symptoms of malaria are nonspecific and similar to those of a minor systemic viral illness. They comprise headache, lassitude, fatigue, abdominal discomfort and muscle and joint aches, usually followed by fever, chills, perspiration, anorexia, vomiting and worsening malaise. In young children, malaria may also present with lethargy, poor feeding and cough. At this early stage of disease progression, with no evidence of vital organ dysfunction, a rapid, full recovery is expected, provided prompt, effective antimalarial treatment is given. If ineffective or poor-quality medicines are given or if treatment is delayed, particularly in *P. falciparum* malaria, the parasite burden often continues to increase and the patient may develop potentially lethal severe malaria. Disease progression to severe malaria may take days but can occur within a few hours.

Severe malaria usually manifests with one or more of the following: coma (cerebral malaria), metabolic acidosis, severe anaemia, hypoglycaemia, acute renal failure or acute pulmonary oedema. If left untreated, severe malaria is fatal in the majority of cases.

2.2 | CLASSIFICATION OF ENDEMICITY

The nature of the clinical disease depends strongly on the background level of acquired protective immunity, which is a consequence of the pattern and intensity of malaria transmission in the area of residence.[4]

Where the transmission of malaria is stable, i.e. where populations are continuously exposed fairly constantly to a high frequency of malarial inoculation (entomological inoculation rate, > 10/year), partial immunity to clinical disease and a reduced risk of developing severe malaria are acquired in early childhood. The pattern of acquired immunity is similar across the sub-Sahel region, where malaria transmission is intense only during the 3- or 4-month rainy season and relatively low at other times. In both these situations, clinical disease is confined mainly to

[4] High transmission area: hyperendemic or holoendemic area in which the prevalence rate of *P. falciparum* parasitaemia among children aged 2–9 years is > 50% most of the year. In these areas, virtually all exposed individuals have been infected by late infancy or early childhood. Moderate transmission area: mesoendemic area in which the prevalence rate of *P. falciparum* parasitaemia among children aged 2–9 years is 11–50% during most of the year. The maximum prevalence of malaria occurs in childhood and adolescence. Low transmission area: hypoendemic area in which the prevalence rate of *P. falciparum* parasitaemia among children aged 2–9 years is ≤ 10% during most of the year. Malaria infection and disease may occur at a similarly low frequency at any age, as little immunity develops.

young children, who may develop high parasite densities that can progress very rapidly to severe malaria. In contrast, in these settings adolescents and adults are partially immune and seldom suffer clinical disease, although they often continue to have low blood-parasite densities. Immunity is modified in pregnancy, and it is gradually lost, at least partially, when individuals move out of the endemic areas for long periods (usually many years).

In areas of unstable malaria transmission, which prevail in much of Asia and Latin America and the remaining parts of the world where malaria is endemic, the intensity of malaria transmission fluctuates widely by season and year and over relatively small distances. *P. vivax* is an important cause of malaria in these regions. The entomological inoculation rate is usually < 5/year and often < 1/year, although there are usually small foci of higher transmission in areas in which asymptomatic parasitaemia is common. The generally low transmission retards acquisition of immunity, so that people of all ages—adults and children alike—suffer from acute clinical malaria, with a significant risk for progression to severe malaria if it is untreated. Epidemics may occur in areas of unstable malaria transmission when the inoculation rate increases rapidly because of a sudden increase in vectorial capacity. Epidemics manifest as a very high incidence of malaria in all age groups and can overwhelm health services. In epidemics, severe malaria is common if prompt, effective treatment is not widely available. Non-immune travellers to a malaria endemic area are at particularly high risk for severe malaria if their infections are not detected promptly and treated effectively.

With effective malaria control, such as population-wide coverage with effective vector control and wide-scale deployment of ACTs, the number of inoculations is usually greatly reduced. This will be followed in time by a corresponding change in the clinical epidemiology of malaria in the area and an increasing risk for an epidemic if control measures are not sustained (see Annex 2).

3RD
EDITION

3 | DIAGNOSIS OF MALARIA

Recommendations on malaria diagnosis

All cases of suspected malaria should have a parasitological test (microscopy or Rapid diagnostic test (RDT)) to confirm the diagnosis.
Both microscopy and RDTs should be supported by a quality assurance programme.
Good practice statement

Prompt, accurate diagnosis of malaria is part of effective disease management. All patients with suspected malaria should be treated on the basis of a confirmed diagnosis by microscopy examination or RDT testing of a blood sample. Correct diagnosis in malaria-endemic areas is particularly important for the most vulnerable population groups, such as young children and non-immune populations, in whom falciparum malaria can be rapidly fatal. High specificity will reduce unnecessary treatment with antimalarial drugs and improve the diagnosis of other febrile illnesses in all settings.

WHO strongly advocates a policy of "test, treat and track" to improve the quality of care and surveillance (section 13).

3.1 | SUSPECTED MALARIA

The signs and symptoms of malaria are non-specific. Malaria is suspected clinically primarily on the basis of fever or a history of fever. There is no combination of signs or symptoms that reliably distinguishes malaria from other causes of fever; diagnosis based only on clinical features has very low specificity and results in overtreatment. Other possible causes of fever and whether alternative or additional treatment is required must always be carefully considered. The focus of malaria diagnosis should be to identify patients who truly have malaria, to guide rational use of antimalarial medicines.

In malaria-endemic areas, malaria should be suspected in any patient presenting with a history of fever or temperature ⩾ 37.5 °C and no other obvious cause. In areas in which malaria transmission is stable (or during the high-transmission period of seasonal malaria), malaria should also be suspected in children with palmar pallor or a haemoglobin concentration of < 8 g/dL. High-transmission settings include many parts of sub-Saharan Africa and some parts of Oceania.

In settings where the incidence of malaria is very low, parasitological diagnosis of all cases of fever may result in considerable expenditure to detect only a few patients with malaria. In these settings, health workers should be trained to identify patients who may have been exposed to malaria (e.g. recent travel to a malaria-endemic area without protective measures) and have fever or a history of fever with no other obvious cause, before they conduct a parasitological test.

In all settings, suspected malaria should be confirmed with a parasitological test. The results of parasitological diagnosis should be available within a short time (< 2 h) of the patient presenting. In settings where parasitological diagnosis is not possible, a decision to provide antimalarial treatment must be based on the probability that the illness is malaria.

In children < 5 years, the practical algorithms for management of the sick child provided by the WHO–United Nations Children's Fund (UNICEF) strategy for Integrated Management of Childhood Illness[5] should be used to ensure full assessment and appropriate case management at first-level health facilities and at the community level.

3.2 | PARASITOLOGICAL DIAGNOSIS

The benefit of parasitological diagnosis relies entirely on an appropriate management response of health care providers. The two methods used routinely for parasitological diagnosis of malaria are light microscopy and immunochromatographic RDTs. The latter detect parasite-specific antigens or enzymes that are either genus or species specific.

Both microscopy and RDTs must be supported by a quality assurance programme. Antimalarial treatment should be limited to cases with positive tests, and patients with negative results should be reassessed for other common causes of fever and treated appropriately.

In nearly all cases of symptomatic malaria, examination of thick and thin blood films by a competent microscopist will reveal malaria parasites. Malaria RDTs should be used if quality-assured malaria microscopy is not readily available. RDTs for detecting *Pf*HRP2 can be useful for patients who have received incomplete antimalarial treatment, in whom blood films can be negative. This is particularly likely if the patient received a recent dose of an artemisinin derivative. If the initial blood film examination is negative in patients with manifestations compatible with severe malaria, a series of blood films should be examined at 6–12-h intervals, or an RDT (preferably one detecting *Pf*HRP2) should be performed. If both the slide examination and the RDT results are negative, malaria is extremely unlikely, and other causes of the illness should be sought and treated.

This document does not include recommendations for use of specific RDTs or for interpreting test results. For guidance, see the WHO manual *Universal access to malaria diagnostic testing*.[6]

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[5] Integrated management of childhood illness for high HIV settings: chart booklet. Geneva, World Health Organization; 2008 (http://www.who.int/child_adolescent_health/documents/9789241597388/en/index.html, accessed 29 October 2009).

[6] Universal access to malaria diagnostic testing—an operational manual. Revised. Geneva: World Health Organization; 2013 (http://www.who.int/entity/malaria/publications/atoz/9789241502092/en/index.html).

In patients with suspected severe malaria and in other high-risk groups, such as patients living with HIV/AIDS, absence or delay of parasitological diagnosis should not delay an immediate start of antimalarial treatment.

At present, molecular diagnostic tools based on nucleic-acid amplification techniques (e.g. loop-mediated isothermal amplification or PCR) do not have a role in the clinical management of malaria.

Where *P. vivax* malaria is common and microscopy is not available, it is recommended that a combination RDT be used that allows detection of *P. vivax* (*p*LDH antigen from *P. vivax)* or pan-malarial antigens (Pan-*p*LDH or aldolase).

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4 | TREATMENT OF UNCOMPLICATED *PLASMODIUM FALCIPARUM* MALARIA

Treating uncomplicated *P. falciparum* malaria

Treatment of uncomplicated P. falciparum malaria

Treat children and adults with uncomplicated *P. falciparum* malaria (except pregnant women in their first trimester) with one of the following recommended ACTs:

- artemether + lumefantrine
- artesunate + amodiaquine
- artesunate + mefloquine
- dihydroartemisinin + piperaquine
- artesunate + sulfadoxine–pyrimethamine (SP).

Strong recommendation, high-quality evidence

Duration of ACT treatment

ACT regimens should provide 3 days' treatment with an artemisinin derivative.

Strong recommendation, high-quality evidence

Revised dose recommendation for dihydroartemisinin + piperaquine in young children

Children weighing <25kg treated with dihydroartemisinin + piperaquine should receive a minimum of 2.5 mg/kg bw per day of dihydroartemisinin and 20 mg/kg bw per day of piperaquine daily for 3 days.

Strong recommendation based on pharmacokinetic modelling

Reducing the transmissibility of treated P. falciparum infections

In low-transmission areas, give a single dose of 0.25 mg/kg bw primaquine with ACT to patients with *P. falciparum* malaria (except pregnant women, infants aged < 6 months and women breastfeeding infants aged < 6 months) to reduce transmission. G6PD testing is not required.

Strong recommendation, low-quality evidence

4.1 | DEFINITION OF UNCOMPLICATED MALARIA

A patient who presents with symptoms of malaria and a positive parasitological test (microscopy or RDT) but with no features of severe malaria is defined as having uncomplicated malaria (see section 7.1 for definition of severe malaria).

4.2 | THERAPEUTIC OBJECTIVES

The clinical objectives of treating uncomplicated malaria are to cure the infection as rapidly as possible and to prevent progression to severe disease. "Cure" is defined as elimination of all parasites from the body. The public health objectives of treatment are to prevent onward transmission of the infection to others and to prevent the emergence and spread of resistance to antimalarial drugs.

4.3 | TREATMENT OF *P. FALCIPARUM* MALARIA

4.3.1 | ARTEMISININ-BASED COMBINATION THERAPY

Treatment of uncomplicated *P. falciparum* malaria

Treat children and adults with uncomplicated *P. falciparum* malaria (except pregnant women in their first trimester) with an ACT.
Strong recommendation, high-quality evidence

GRADE (Annex 4, A4.1 and A4.2)
In the absence of resistance to the partner drug, the five recommended ACTs have all been shown to achieve a PCR - adjusted treatment failure rate of < 5% in many trials in several settings in both adults and children (*high-quality evidence*).

Other considerations
The guideline development group decided to recommend a menu of approved combinations, from which countries can select first- and second-line treatment.

Sinclair D, Zani B, Donegan S, Olliaro PL, Garner P. Artemisinin-based combination therapy for treating uncomplicated malaria. Cochrane Database Systemat Rev 2009; 3:CD007483. doi: 10.1002/14651858.CD007483.pub2.

Zani B, Gathu M, Donegan S, Olliaro PL, Sinclair D. Dihydroartemisinin-piperaquine for treating uncomplicated Plasmodium falciparum malaria. Cochrane Database of Systemat Rev 2014;1:CD010927. doi: 10.1002/14651858.CD010927.

ACT is a combination of a rapidly acting artemisinin derivative with a longer-acting (more slowly eliminated) partner drug. The artemisinin component rapidly clears parasites from the blood (reducing parasite numbers by a factor of approximately 10 000 in each 48-h asexual cycle) and is also active against the sexual stages of parasite that mediate onward transmission to mosquitos. The longer-acting partner drug clears the remaining parasites and provides protection against development of resistance to the artemisinin derivative. Partner drugs with longer elimination half-lives also provide a period of post-treatment prophylaxis.

The five ACTs recommended for treatment of uncomplicated *P. falciparum* malaria are:

- artemether + lumefantrine
- artesunate + amodiaquine
- artesunate + mefloquine
- artesunate + SP
- dihydroartemisinin + piperaquine.

4.3.2 | DURATION OF TREATMENT

Duration of ACT treatment

ACT regimens should provide 3 days' treatment with an artemisinin-derivative.
Strong recommendation, high-quality evidence

GRADE (see Annex 4, A4.3)
In four randomized controlled trials in which the addition of 3 days of artesunate to SP was compared directly with 1 day of artesunate with SP:

- Three days of artesunate reduced the PCR-adjusted treatment failure rate within the first 28 days from that with 1 day of artesunate (RR, 0.45; 95% CI, 0.36–0.55, four trials, 1202 participants, *high-quality evidence*).
- Three days of artesunate reduced the number of participants who had gametocytaemia at day 7 from that with 1 day of artesunate (RR, 0.74; 95% CI, 0.58–0.93, four trials, 1260 participants, *high-quality evidence*).

Other considerations
The guideline development group considered that 3 days of artemisinin derivative are necessary to provide sufficient efficacy, promote good adherence and minimize the risk of drug resistance resulting from incomplete treatment.

A 3-day course of the artemisinin component of ACTs covers two asexual cycles, ensuring that only a small fraction of parasites remain for clearance by the partner drug, thus reducing the potential development of resistance to the partner drug. Shorter courses (1–2 days) are therefore not recommended, as they are less effective, have less effect on gametocytes and provide less protection for the slowly eliminated partner drug.

4.3.3 | DOSING OF ACTS

ACT regimens must ensure optimal dosing to prolong their useful therapeutic life, i.e. to maximize the likelihood of rapid clinical and parasitological cure, minimize transmission and retard drug resistance.

It is essential to achieve effective antimalarial drug concentrations for a sufficient time (exposure) in all target populations in order to ensure high cure rates. The dosage recommendations below are derived from understanding the relationship between dose and the profiles of exposure to the drug (pharmacokinetics) and the resulting therapeutic efficacy (pharmacodynamics) and safety. Some patient groups, notably younger children, are not dosed optimally with the "dosage regimens recommended by manufacturers, which compromises efficacy and fuels resistance. In these guidelines when there was pharmacological evidence that certain patient groups are not receiving optimal doses, dose regimens were adjusted to ensure similar exposure across all patient groups.

Weight-based dosage recommendations are summarized below. While age-based dosing may be more practical in children, the relation between age and weight differs in different populations. Age-based dosing can therefore result in under-dosing or over-dosing of some patients, unless large, region-specific weight-for-age databases are available to guide dosing in that region.

Factors other than dosage regimen may also effect exposure to a drug and thus treatment efficacy. The drug exposure of an individual patient also depends on factors such as the quality of the drug, the formulation, adherence and, for some drugs, co-administration with fat. Poor adherence is a major cause of treatment failure and drives the emergence and spread of drug resistance. Fixed-dose combinations encourage adherence and are preferred to loose (individual) tablets. Prescribers should take the time necessary to explain to patients why they should complete antimalarial course.

Artemether + lumefantrine

(For further details, see Annex 5, A5.3.)

Formulations currently available: Dispersible or standard tablets containing 20 mg artemether and 120 mg lumefantrine, and standard tablets containing 40 mg artemether and 240 mg lumefantrine in a fixed-dose combination formulation. The flavoured dispersible tablet paediatric formulation facilitates use in young children.

Target dose range: A total dose of 5–24 mg/kg bw of artemether and 29–144 mg/kg bw of lumefantrine

Recommended dosage regimen: Artemether + lumefantrine is given twice a day for 3 days (total, six doses). The first two doses should, ideally, be given 8 h apart.

Body weight (kg)	Dose (mg) of artemether + lumefantrine given twice daily for 3 days
5 to < 15	20 + 120
15 to < 25	40 + 240
25 to < 35	60 + 360
⩾ 35	80 + 480

Factors associated with altered drug exposure and treatment response:

- Decreased exposure to lumefantrine has been documented in young children (<3 years)as well as pregnant women, large adults, patients taking mefloquine, rifampicin or efavirenz and in smokers. As these target populations may be at increased risk for treatment failure, their responses to treatment should be monitored more closely and their full adherence ensured.
- Increased exposure to lumefantrine has been observed in patients concomitantly taking lopinavir- lopinavir/ritonavir-based antiretroviral agents but with no increase in toxicity; therefore, no dosage adjustment is indicated.

Additional comments:

- An advantage of this ACT is that lumefantrine is not available as a monotherapy and has never been used alone for the treatment of malaria.
- Absorption of lumefantrine is enhanced by co-administration with fat. Patients or caregivers should be informed that this ACT should be taken immediately after food or a fat containing drink (e.g. milk), particularly on the second and third days of treatment.

Artesunate + amodiaquine

(For further details see Annex 5, A5.1 and A5.4.)

Formulations currently available: A fixed-dose combination in tablets containing 25 + 67.5 mg, 50 + 135 mg or 100 + 270 mg of artesunate and amodiaquine, respectively

Target dose and range: The target dose (and range) are 4 (2–10) mg/kg bw per day artesunate and 10 (7.5–15) mg/kg bw per day amodiaquine once a day for 3 days. A total therapeutic dose range of 6–30 mg/kg bw per day artesunate and 22.5–45 mg/kg bw per dose amodiaquine is recommended.

Body weight (kg)	Artesunate + amodiaquine dose (mg) given daily for 3 days
4.5 to < 9	25 + 67.5
9 to < 18	50 + 135
18 to < 36	100 + 270
⩾ 36	200 + 540

Factors associated with altered drug exposure and treatment response:

- Treatment failure after amodiaquine monotherapy was more frequent among children who were underweight for their age. Therefore, their response to artesunate + amodiaquine treatment should be closely monitored.
- Artesunate + amodiaquine is associated with severe neutropenia, particularly in patients co-infected with HIV and especially in those on zidovudine and/or cotrimoxazole. Concomitant use of efavirenz increases exposure to amodiaquine and hepatotoxicity. Thus, concomitant use of artesunate + amodiaquine by patients taking zidovudine, efavirenz and cotrimoxazole should be avoided, unless this is the only ACT promptly available.

Additional comments:

- No significant changes in the pharmacokinetics of amodiaquine or its metabolite desethylamodiaquine have been observed during the second and third trimesters of pregnancy; therefore, no dosage adjustments are recommended.
- No effect of age has been observed on the plasma concentrations of amodiaquine and desethylamodiaquine, so no dose adjustment by age is indicated. Few data are available on the pharmacokinetics of amodiaquine in the first year of life.

Artesunate + mefloquine

(For further details, see Annex 5, A5.4 and A5.10.)

Formulations currently available: A fixed-dose formulation of paediatric tablets containing 25 mg artesunate and 55 mg mefloquine hydrochloride (equivalent to 50 mg mefloquine base) and adult tablets containing 100 mg artesunate and 220 mg mefloquine hydrochloride (equivalent to 200 mg mefloquine base)

Target dose and range: Target doses (ranges) of 4 (2–10) mg/kg bw per day artesunate and 8.3 (5–11) mg/kg bw per day mefloquine, given once a day for 3 days

Body weight (kg)	Artesunate + mefloquine dose (mg) given daily for 3 days
5 to < 9	25 + 55
9 to < 18	50 + 110
18 to < 30	100 + 220
⩾ 30	200 + 440

Additional comments:

- Mefloquine was associated with increased incidences of nausea, vomiting, dizziness, dysphoria and sleep disturbance in clinical trials, but these symptoms are seldom debilitating, and, where this ACT has been used, it has generally been well tolerated. To reduce acute vomiting and optimize absorption, the total mefloquine dose should preferably be split over 3 days, as in current fixed-dose combinations.

- As concomitant use of rifampicin decreases exposure to mefloquine, potentially decreasing its efficacy, patients taking this drug should be followed up carefully to identify treatment failures.

Artesunate + sulfadoxine–pyrimethamine

(For further details, see Annex 5, A5.4 and A5.13.)

Formulations: Currently available as blister-packed, scored tablets containing 50 mg artesunate and fixed dose combination tablets comprising 500 mg sulfadoxine + 25 mg pyrimethamine. There is no fixed-dose combination.

Target dose and range: A target dose (range) of 4 (2–10) mg/kg bw per day artesunate given once a day for 3 days and a single administration of at least 25 / 1.25 (25–70 / 1.25–3.5) mg/kg bw sulfadoxine / pyrimethamine given as a single dose on day 1

Body weight (kg)	Artesunate dose given daily for 3 days (mg)	Sulfadoxine / pyrimethamine dose (mg) given as a single dose on day 1
5 to < 10	25 mg	250 / 12.5
10 to < 25	50 mg	500 / 25
25 to < 50	100 mg	1000 / 50
⩾ 50	200 mg	1500 / 75

Factors associated with altered drug exposure and treatment response: The low dose of folic acid (0.4 mg daily) that is required to protect the fetuses of pregnant women from neural tube defects do not reduce the efficacy of SP, whereas higher doses (5 mg daily) do significantly reduce its efficacy and should not be given concomitantly.

Additional comments:

- The disadvantage of this ACT is that it is not available as a fixed-dose combination. This may compromise adherence and increase the risk for distribution of loose artesunate tablets, despite the WHO ban on artesunate monotherapy.
- Resistance is likely to increase with continued widespread use of SP, sulfalene–pyrimethamine and cotrimoxazole (trimethoprim-sulfamethoxazole). Fortunately, molecular markers of resistance to antifols and sulfonamides correlate well with therapeutic responses. These should be monitored in areas in which this drug is used.

Dihydroartemisinin + piperaquine

(For further details, see Annex 5, A5.8.)

Revised dose recommendation for dihydroartemisinin-piperaquine in young children

Children weighing <25kg treated with dihydroarteminsinin + piperaquine should receive a minimum of 2.5 mg/kg bw per day of dihydroartemisinin and 20 mg/kg bw per day of piperaquine daily for 3 days.
Strong recommendation based on pharmacokinetic modelling

The dosing subgroup reviewed clinical efficacy and safety data together with all available dihydroartemisinin-piperaquine pharmacokinetic data (6 published studies and 10 studies from the WWARN database; total 652 patients) and then conducted simulations of piperaquine exposures for each weight group. These showed lower exposure in younger children with higher risks of treatment failure. The revised dose regimens are predicted to provide equivalent piperaquine exposures across all age groups. The subgroup also reviewed preliminary results from an unpublished study using doses similar to those now recommended in this guidelines (n=100).

Other considerations
This dose adjustment is not predicted to result in higher peak piperaquine concentrations than in older children and adults, and as there is no evidence of increased toxicity in young children, the GRC concluded that the predicted benefits of improved antimalarial exposure are not at the expense of increased risk.

Tarning J, Zongo I, Some FA, Rouamba N, Parikh S, Rosenthal PJ, et al. Population pharmacokinetics and pharmacodynamics of piperaquine in children with uncomplicated falciparum malaria. Clin Pharmacol Ther 2012;91:497–505.

The WorldWide Antimalarial Resistance Network DP Study Group. The effect of dosing regimens on the antimalarial efficacy of dihydroartemisinin–piperaquine: a pooled analysis of individual patient data. PLoS Med 2013;10: e1001564.

Formulations: Currently available as a fixed-dose combination in tablets containing 40 mg dihydroartemisinin and 320 mg piperaquine and paediatric tablets contain 20 mg dihydroartemisinin and 160 mg piperaquine.

Target dose and range: A target dose (range) of 4 (2–10) mg/kg bw per day dihydroartemisinin and 18 (16–27) mg/kg bw per day piperaquine given once a day for 3 days for adults and children weighing ⩾ 25 kg. The target doses and ranges for children weighing < 25 kg are 4 (2.5–10) mg/kg bw per day dihydroartemisinin and 24 (20–32) mg/kg bw per day piperaquine once a day for 3 days.

Recommended dosage regimen: The dose regimen currently recommended by the manufacturer provides adequate exposure to piperaquine and excellent cure rates (> 95%), except in children < 5 years, who have a threefold increased risk for treatment failure. Children in this age group have significantly lower plasma piperaquine concentrations than older children and adults given the same mg/kg bw dose. Children weighing < 25 kg should receive at least 2.5 mg/kg bw dihydroartemisinin and 20 mg/kg bw piperaquine to achieve the same exposure as children weighing ⩾ 25 kg and adults.

Dihydroartemisinin + piperaquine should be given daily for 3 days.

Body weight (kg)	Dihydroartemisinin + piperaquine dose (mg) given daily for 3 days
5 to < 8	20 + 160
8 to < 11	30 + 240
11 to < 17	40 + 320
17 to < 25	60 + 480
25 to < 36	80 + 640
36 to < 60	120 + 960
60 < 80	160 + 1280
>80	200 + 1600

Factors associated with altered drug exposure and treatment response:

- High-fat meals should be avoided, as they significantly accelerate the absorption of piperaquine, thereby increasing the risk for potentially arrhythmogenic delayed ventricular repolarization (prolongation of the corrected electrocardiogram QT interval). Normal meals do not substantially alter the absorption of piperaquine.
- As malnourished children are at increased risk for treatment failure, their response to treatment should be monitored closely.
- Dihydroartemisinin exposure is lower in pregnant women.
- Piperaquine is eliminated more rapidly by pregnant women, shortening the post-treatment prophylactic effect of dihydroartemisinin + piperaquine. As this does not affect primary efficacy, no dosage adjustment is recommended for pregnant women.

Additional comments: Piperaquine prolongs the QT interval by approximately the same amount as chloroquine but by less than quinine. It is not necessary to perform an electrocardiogram before prescribing dihydroartemisinin + piperaquine, but this ACT should not be used in patients with congenital QT prolongation or who have a clinical condition or are on medications (see Annex 5.14) that prolong the QT interval. There has been no evidence of piperaquine-related cardiotoxicity in large randomized trials or in extensive deployment.

4.4 | RECURRENT FALCIPARUM MALARIA

Recurrence of *P. falciparum* malaria can result from re-infection or recrudescence (treatment failure). Treatment failure may result from drug resistance or inadequate exposure to the drug due to sub-optimal dosing, poor adherence, vomiting, unusual pharmacokinetics in an individual or substandard medicines. It is important to determine from the patient's history whether he or she vomited the previous treatment or did not complete a full course of treatment.

When possible, treatment failure must be confirmed parasitologically. This may require referring the patient to a facility with microscopy or LDH-based RDTs, as *P. falciparum* histidine-rich protein-2 (*Pf*HRP2)-based tests may remain positive for weeks after the initial infection, even without recrudescence. Referral may be necessary anyway to obtain second-line treatment. In individual patients, it may not be possible to distinguish recrudescence from re-infection, although lack of resolution of fever and parasitaemia or their recurrence within 4 weeks of treatment are considered failures of treatment with currently recommended ACTs. In many cases, treatment failures are missed because patients are not asked whether they received antimalarial treatment within the preceding 1–2 months. Patients who present with malaria should be asked this question routinely.

4.4.1 | FAILURE WITHIN 28 DAYS

The recommended second-line treatment is an alternative ACT known to be effective in the region. Adherence to 7-day treatment regimens (with artesunate or quinine both of which should be co-administered with + tetracycline, or doxycycline or clindamycin) is likely to be poor if treatment is not directly observed; these regimens are no longer generally recommended. The distribution and use of oral artesunate monotherapy outside special centres is strongly discouraged, and quinine-containing regimens are not well tolerated.

4.4.2 | FAILURE AFTER 28 DAYS

Recurrence of fever and parasitaemia > 4 weeks after treatment may be due to either recrudescence or a new infection. The distinction can be made only by PCR genotyping of parasites from the initial and the recurrent infections.

As PCR is not routinely used in patient management, all presumed treatment failures after 4 weeks of initial treatment should, from an operational standpoint, be considered new infections and be treated with the first-line ACT. However, reuse of mefloquine within 60 days of first treatment is associated with an increased risk for neuropsychiatric reactions, and an alternative ACT should be used.

4.5 | REDUCING THE TRANSMISSIBILITY OF TREATED *P. FALCIPARUM* INFECTIONS IN AREAS OF LOW-INTENSITY TRANSMISSION

Reducing the transmissibility of *P. falciparum* infections

In low-transmission areas, give a single dose of 0.25 mg/kg bw primaquine with ACT to patients with *P. falciparum* malaria (except pregnant women, infants aged < 6 months and women breastfeeding infants aged < 6 months) to reduce transmission. G6PD testing is not required.
Strong recommendation, low-quality evidence

GRADE (see Annex 4, A4.4)
In an analysis of observational studies of single-dose primaquine, data from mosquito feeding studies on 180 people suggest that adding 0.25 mg/kg primaquine to treatment with an ACT can rapidly reduce the infectivity of gametocytes to mosquitoes.

In a systematic review of eight randomized controlled trials of the efficacy of adding single-dose primaquine to ACTs for reducing the transmission of malaria, in comparison with ACTs alone:

- Single doses of > 0.4 mg/kg bw primaquine reduced gametocyte carriage at day 8 by about two thirds (RR, 0.34; 95% CI, 0.19–0.59, two trials, 269 participants, *high-quality evidence*); and
- Single doses of primaquine > 0.6 mg/kg bw reduced gametocyte carriage at day 8 by about two thirds (RR, 0.29; 95% CI, 0.22–0.37, seven trials, 1380 participants, *high-quality evidence*).

There have been no randomized controlled trials of the effects on the incidence of malaria or on transmission to mosquitos.

Other considerations
The guideline development group considered that the evidence of a dose–response relation from observational studies of mosquito feeding was sufficient to conclude the primaquine dose of 0.25mg/kg bw significantly reduced *P .falciparum* transmissibility.

Graves PM, Gelband H, Garner P. Primaquine or other 8-aminoquinoline for reducing P. falciparum transmission. Cochrane Database Systemat Rev 2014; 6:CD008152. doi: 10.1002/14651858.CD008152.pub3.

White NJ, Qiao LG, Qi G, Luzzatto L. Rationale for recommending a lower dose of primaquine as a Plasmodium falciparum gametocytocide in populations where G6PD deficiency is common. Malar J 2012;11:418.

The population benefits of reducing malaria transmission with gametocytocidal drugs such as primaquine require that a very high proportion of treated patients receive these medicines and that there is no large transmission reservoir of asymptomatic parasite carriers. This strategy is therefore likely to be effective only in areas of low-intensity malaria transmission, as a component of pre-elimination or elimination programmes.

In light of concern about the safety of the previously recommended dose of 0.75 mg/kg bw in individuals with G6PD deficiency, a WHO panel reviewed the safety of primaquine as a *P. falciparum* gametocytocide and concluded that a single dose of 0.25 mg/kg bw of primaquine base is unlikely to cause serious toxicity, even in people with G6PD deficiency.[7] Thus, where indicated a single dose of 0.25mg/kg bw of primaquine base should be given on the first day of treatment, in addition to an ACT, to all patients with parasitologically confirmed *P.falciparum* malaria except for pregnant women, infants < 6 months of age and women breastfeeding infants < 6 months of age, because there are insufficient data on the safety of its use in these groups. (For further details, see Annex 5, A5.11.)

Dosing table based on the most widely currently available tablet strength (7.5mg base)

Body weight (kg)	Single dose of primaquine (mg base)
10[a] to < 25	3.75
25 to < 50	7.5
50 to 100	15

[a] Dosing of young children weighing < 10 kg is limited by the tablet sizes currently available.

4.6 | INCORRECT APPROACHES TO TREATMENT

4.6.1 | USE OF MONOTHERAPY

The continued use of artemisinins or any of the partner medicines alone will compromise the value of ACT by selecting for drug resistance.

As certain patient groups, such as pregnant women, may need specifically tailored combination regimens, single artemisinin derivatives will still be used in selected referral facilities in the public sector, but they should be withdrawn entirely from the private and informal sectors and from peripheral public health care facilities.

[7] Recht J, Ashley E, White N. Safety of 8-aminoquinoline antimalarial medicines. Geneva: World Health Organization; 2014 (http://who.int/malaria/publications/atoz/9789241506977/en/).

Similarly, continued availability of amodiaquine, mefloquine and SP as monotherapies in many countries is expected to shorten their useful therapeutic life as partner drugs of ACT, and they should be withdrawn wherever possible.

4.6.2 | INCOMPLETE DOSING

In endemic regions, some semi-immune malaria patients are cured by an incomplete course of antimalarial drugs or by a treatment regimen that would be ineffective in patients with no immunity. In the past, this led to different recommendations for patients considered semi-immune and those considered non-immune. As individual immunity can vary considerably, even in areas of moderate-to-high transmission intensity, this practice is no longer recommended. A full treatment course with a highly effective ACT is required whether or not the patient is considered to be semi-immune.

Another potentially dangerous practice is to give only the first dose of a treatment course to patients with suspected but unconfirmed malaria, with the intention of giving the full treatment if the diagnosis is confirmed. This practice is unsafe, could engender resistance, and is not recommended.

4.7 | ADDITIONAL CONSIDERATIONS FOR CLINICAL MANAGEMENT

4.7.1 | CAN THE PATIENT TAKE ORAL MEDICATION?

Some patients cannot tolerate oral treatment and will require parenteral or rectal administration for 1–2 days, until they can swallow and retain oral medication reliably. Although such patients do not show other signs of severity, they should receive the same initial antimalarial treatments recommended for severe malaria (see chapter 7, 7.4.). Initial rectal or parenteral treatment must always be followed by a full 3-day course of ACT.

4.7.2 | USE OF ANTIPYRETICS

In young children, high fevers are often associated with vomiting, regurgitation of medication and seizures. They are thus treated with antipyretics and, if necessary, fanning and tepid sponging. Antipyretics should be used if the core temperature is > 38.5 °C. Paracetamol (acetaminophen) at a dose of 15 mg/kg bw every 4 h is widely used; it is safe and well tolerated and can be given orally or as a suppository. Ibuprofen (5 mg/kg bw) has been used successfully as an alternative in the treatment of malaria and other childhood fevers, but, like aspirin and other non-steroidal anti-inflammatory drugs, it is *no longer recommended* because of the risks of gastrointestinal bleeding, renal impairment and Reye's syndrome.

4.7.3 | USE OF ANTI-EMETICS

Vomiting is common in acute malaria and may be severe. Parenteral antimalarial treatment may therefore be required until oral administration is tolerated. Then a full 3-day course of ACT should be given. Anti-emetics are potentially sedative and may have neuropsychiatric adverse effects, which could mask or confound the diagnosis of severe malaria. They should therefore be used with caution.

4.7.4 | MANAGEMENT OF SEIZURES

Generalized seizures are more common in children with *P. falciparum* malaria than in those with malaria due to other species. This suggests an overlap between the cerebral pathology resulting from falciparum malaria and febrile convulsions. As seizures may be a prodrome of cerebral malaria, patients who have more than two seizures within a 24-h period should be treated as for severe malaria. If the seizures continue, the airways should be maintained and anticonvulsants given (parenteral or rectal benzodiazepines or intramuscular paraldehyde). When the seizure has stopped, the child should be treated as indicated in section 7.10.5, if his or her core temperature is > 38.5 °C. There is no evidence that prophylactic anticonvulsants are beneficial in otherwise uncomplicated malaria, and they are not recommended.

3RD EDITION

5 | TREATMENT OF UNCOMPLICATED *P. FALCIPARUM* MALARIA IN SPECIAL RISK GROUPS

Treating uncomplicated *P. falciparum* malaria in special risk groups

First trimester of pregnancy

Treat pregnant women with uncomplicated *P. falciparum* malaria during the first trimester with 7 days of quinine + clindamycin.
Strong recommendation, very low- quality evidence

Infants less than 5kg body weight

Treat infants weighing < 5 kg with uncomplicated *P. falciparum* malaria with an ACT at the same mg/kg bw target dose as for children weighing 5 kg.
Strong recommendation, very low- quality evidence

Patients co-infected with HIV

In people who have HIV/AIDS and uncomplicated *P. falciparum* malaria, avoid artesunate + SP if they are also receiving co-trimoxazole, and avoid artesunate + amodiaquine if they are also receiving efavirenz or zidovudine.
Good practice statement

Non-immune travellers

Treat travellers with uncomplicated *P. falciparum* malaria returning to non-endemic settings with an ACT.
Strong recommendation, high-quality evidence

Uncomplicated hyperparasitaemia

People with *P. falciparum* hyperparasitaemia are at increased risk of treatment failure, severe malaria and death so should be closely monitored, in addition to receiving an ACT.
Good practice statement

Several important patient sub-populations, including young children, pregnant women and patients taking potent enzyme inducers (e.g. rifampicin, efavirenz), have altered pharmacokinetics, resulting in sub-optimal exposure to antimalarial drugs. This increases the rate of treatment failure with current dosage regimens. The rates of treatment failure are substantially higher in hyperparasitaemic patients and patients in areas with artemisinin-resistant falciparum malaria, and these groups require greater exposure to antimalarial drugs (longer duration of therapeutic concentrations) than is achieved with current ACT dosage recommendations. It is often uncertain how best to achieve this. Options include increasing individual doses, increasing the frequency or duration of dosing, or adding an additional antimalarial drug. Increasing individual doses may not, however, achieve the desired exposure (e.g. lumefantrine absorption becomes saturated), or the dose may be toxic due to transiently high plasma concentrations (piperaquine, mefloquine, amodiaquine, pyronaridine). An additional advantage of lengthening the duration of treatment (by giving a 5-day regimen) is that it provides additional exposure of the asexual cycle to the artemisinin component as well as augmenting exposure to the partner drug. The acceptability, tolerability, safety and effectiveness of augmented ACT regimens in these special circumstances should be evaluated urgently.

5.1 | PREGNANT AND LACTATING WOMEN

Malaria in pregnancy is associated with low-birth-weight infants, increased anaemia and, in low-transmission areas, increased risks for severe malaria, pregnancy loss and death. In high-transmission settings, despite the adverse effects on fetal growth, malaria is usually asymptomatic in pregnancy or is associated with only mild, non-specific symptoms. There is insufficient information on the safety, efficacy and pharmacokinetics of most antimalarial agents in pregnancy, particularly during the first trimester.

5.1.1 | FIRST TRIMESTER

First trimester of pregnancy

Treat pregnant women with uncomplicated *P. falciparum* malaria during the first trimester with 7 days of quinine + clindamycin.
Strong recommendation

Evidence supporting the recommendation (see Annex 4, A4.5)
Data available were not suitable for evaluation using the GRADE methodology, as there is no /almost no evidence for alternative treatment using ACT.

Safety assessment from published prospective data on 700 women exposed in the first trimester of pregnancy has not indicated any adverse effects of artemisinin-derivatives on pregnancy or on the health of the fetus or neonate.

The currently available data are only sufficient to exclude a ⩾ 4.2-fold increase in risk of any major defect detectable at birth (background prevalence assumed to be 0.9%), if half the exposures occur during the embryo-sensitive period (4–9 weeks post-conception).

Other considerations
The limited data available on the safety of artemisinin-derivatives in early pregnancy allow for some reassurance in counselling women accidentally exposed to an artemisinin-derivative early in the first trimester. There is no need for them to have their pregnancy interrupted because of this exposure.

In the absence of adequate safety data on the artemisinin-derivatives in the first trimester of pregnancy the Guideline Development Group was unable to make recommendations beyond reiterating the status quo.

McGready R, Lee SJ, Wiladphaingern J, et al. Adverse effects of falciparum and vivax malaria and the safety of antimalarial treatment in early pregnancy: a population-based study. Lancet Infect Dis. 2012;12:388-96.

Mosha D, Mazuguni F, Mrema S, et al. Safety of artemether-lumefantrine exposure in first trimester of pregnancy: an observational cohort. Malar J. 2014; 13: 197e.

Because organogenesis occurs mainly in the first trimester, this is the time of greatest concern for potential teratogenicity, although development of the nervous system continues throughout pregnancy. The antimalarial medicines considered safe in the first trimester of pregnancy are quinine, chloroquine, clindamycin and proguanil.

The safest treatment regimen for pregnant women in the first trimester with uncomplicated falciparum malaria is therefore quinine + clindamycin (10mg/kg bw twice a day) for 7 days (or quinine monotherapy if clindamycin is not available). An ACT or oral artesunate + clindamycin is an alternative if quinine + clindamycin is not available or fails.

In reality, women often do not declare their pregnancy in the first trimester or may not yet be aware that they are pregnant. Therefore, all women of childbearing age should be asked about the possibility that they are pregnant before they are given antimalarial agents; this is standard practice for administering any medicine to potentially pregnant women. Nevertheless, women in early pregnancy will often be exposed inadvertently to the available first-line treatment, mostly ACT. Published prospective data on 700 women exposed in the first trimester of pregnancy indicate no adverse effects of artemisinins (or the partner drugs) on pregnancy or on the health of fetuses or neonates. The available data are sufficient to exclude a ⩾ 4.2-fold increase in risk of any major defect detectable at birth (background prevalence assumed to be 0.9%), if half the exposures occur during the embryo-sensitive period (4–9 weeks post-conception). These data provide assurance in counselling women exposed to an antimalarial drug early in the first trimester and indicate that there is no need for them to have their pregnancy interrupted because of this exposure.

5.1.2 | SECOND AND THIRD TRIMESTERS

Experience with artemisinin derivatives in the second and third trimesters (over 4000 documented pregnancies) is increasingly reassuring: no adverse effects on the mother or fetus have been reported. The current assessment of risk–benefit suggests that *ACTs should be used* to treat uncomplicated falciparum malaria in the second and third trimesters of pregnancy. The current standard six-dose artemether + lumefantrine regimen for the treatment of uncomplicated falciparum malaria has been evaluated in > 1000 women in the second and third trimesters in controlled trials and has been found to be well tolerated and safe. In a low-transmission setting on the Myanmar–Thailand border, however, the efficacy of the standard six-dose artemether + lumefantrine regimen was inferior to 7 days of artesunate monotherapy. The lower efficacy may have been due to lower drug concentrations in pregnancy, as was also recently observed in a high-transmission area in Uganda and the United Republic of Tanzania. Although many women in the second and third trimesters of pregnancy in Africa have been exposed to artemether + lumefantrine, further studies are under way to evaluate its efficacy, pharmacokinetics and safety in pregnant women. Similarly, many pregnant women in Africa have been treated with amodiaquine alone or combined with SP or artesunate;

however, amodiaquine use for the treatment of malaria in pregnancy has been formally documented in only > 1300 pregnancies. Use of amodiaquine in women in Ghana in the second and third trimesters of pregnancy was associated with frequent minor side-effects but not with liver toxicity, bone marrow depression or adverse neonatal outcomes.

Dihydroartemisinin + piperaquine was used successfully in the second and third trimesters of pregnancy in > 2000 women on the Myanmar–Thailand border for rescue therapy and in Indonesia for first-line treatment. SP, although considered safe, is not appropriate for use as an artesunate partner drug in many areas because of resistance to SP. If artesunate + SP is used for treatment, co-administration of daily high doses (5 mg) of folate supplementation should be avoided, as this compromises the efficacy of SP. A lower dose of folate (0.4–0.5 mg bw/day) or a treatment other than artesunate + SP should be used.

Mefloquine is considered safe for the treatment of malaria during the second and third trimesters; however, it should be given only in combination with an artemisinin derivative.

Quinine is associated with an increased risk for hypoglycaemia in late pregnancy, and it should be used (with clindamycin) only if effective alternatives are not available.

Primaquine and tetracyclines should not be used in pregnancy.

5.1.3 | DOSING IN PREGNANCY

Data on the pharmacokinetics of antimalarial agents used during pregnancy are limited. Those available indicate that pharmacokinetic properties are often altered during pregnancy but that the alterations are insufficient to warrant dose modifications at this time. With quinine, no significant differences in exposure have been seen during pregnancy. Studies of the pharmacokinetics of SP used in IPTp in many sites show significantly decreased exposure to sulfadoxine, but the findings on exposure to pyrimethamine are inconsistent. Therefore, no dose modification is warranted at this time.

Studies are available of the pharmacokinetics of artemether + lumefantrine, artesunate + mefloquine and dihydroartemisinin + piperaquine. Most data exist for artemether + lumefantrine; these suggest decreased overall exposure during the second and third trimesters. Simulations suggest that a standard six-dose regimen of lumefantrine given over 5 days, rather than 3 days, improves exposure, but the data are insufficient to recommend this alternative regimen at present. Limited data on pregnant women treated with dihydroartemesinin + piperaquine suggest lower dihydroartemisinin exposure and no overall difference in total piperaquine exposure, but a shortened piperaquine elimination half-life was noted. The data on artesunate + mefloquine are insufficient to recommend an adjustment of dosage. No data are available on the pharmacokinetics of artesunate + amodiaquine in pregnant women with falciparum malaria, although drug exposure was similar in pregnant and non-pregnant women with vivax malaria.

5.1.4 | LACTATING WOMEN

The amounts of antimalarial drugs that enter breast milk and are consumed by breastfeeding infants are relatively small. Tetracycline is contraindicated in breastfeeding mothers because of its potential effect on infants' bones and teeth. Pending further information on excretion in breast milk, primaquine should not be used for nursing women, unless the breastfed infant has been checked for G6PD deficiency.

5.2 | YOUNG CHILDREN AND INFANTS (INCLUDING THOSE WHO ARE MALNOURISHED)

Artemisinin derivatives are safe and well tolerated by young children; therefore, the choice of ACT is determined largely by the safety and tolerability of the partner drug.

SP (with artesunate) should be avoided in the first weeks of life because it displaces bilirubin competitively and could thus aggravate neonatal hyperbilibinaemia. Primaquine should be avoided in the first 6 months of life (although there are no data on its toxicity in infants), and tetracyclines should be avoided throughout infancy. With these exceptions, none of the other currently recommended antimalarial treatments has shown serious toxicity in infancy.

Delay in treating *P. falciparum* malaria in infants and young children can have fatal consequences, particularly for more severe infections. The uncertainties noted above should not delay treatment with the most effective drugs available. In treating young children, it is important to ensure accurate dosing and retention of the administered dose, as infants are more likely to vomit or regurgitate antimalarial treatment than older children or adults. Taste, volume, consistency and gastrointestinal tolerability are important determinants of whether the child retains the treatment. Mothers often need advice on techniques of drug administration and the importance of administering the drug again if it is regurgitated within 1 h of administration. Because deterioration in infants can be rapid, the threshold for use of parenteral treatment should be much lower.

5.2.1 | OPTIMAL ANTIMALARIAL DOSING IN YOUNG CHILDREN

Although dosing on the basis of body area is recommended for many drugs in young children, for the sake of simplicity, antimalarial drugs have been administered as a standard dose per kg bw for all patients, including young children and infants. This approach does not take into account changes in drug disposition that occur with development. The currently recommended doses of lumefantrine, piperaquine, SP, artesunate and chloroquine result in lower drug concentrations in young children and infants than in older patients. Adjustments to previous dosing regimens for dihydroartemisinin + piperaquine in uncomplicated malaria and for artesunate in severe malaria are now recommended to ensure adequate the drug exposure in this vulnerable population. The available evidence for artemether + lumefantrine, SP and chloroquine does not indicate dose modification at this time, but young children should be closely monitored, as reduced drug exposure may increase the risk for treatment failure. Limited studies of amodiaquine and mefloquine showed no significant effect of age on plasma concentration profiles.

In community situations where parenteral treatment is needed but cannot be given, such as for infants and young children who vomit antimalarial drugs repeatedly or are too weak to swallow or are very ill, give rectal artesunate and transfer the patient to a facility in which parenteral treatment is possible. Rectal administration of a single dose of artesunate as pre-referral treatment reduces the risks for death and neurological disability, as long as this initial treatment is followed by appropriate parenteral antimalarial treatment in hospital. Further evidence on pre-referral rectal administration of artesunate and other antimalarial drugs is given in section 7.5.

5.2.2 | OPTIMAL ANTIMALARIAL DOSING IN INFANTS

Infants less than 5kg body weight

Treat infants weighing < 5 kg with uncomplicated *P. falciparum* malaria with an ACT at the same mg/kg bw target dose as for children weighing 5 kg.
Strong recommendation

Evidence supporting the recommendation (see Annex 4, A4.6)
Data available were not suitable for evaluation using the GRADE methodology.

In most clinical studies, subgroups of infants and older children were not distinguished, and the evidence for young infants (< 5 kg) is insufficient for confidence in current treatment recommendations. Nevertheless despites these uncertainties, infants need prompt, effective treatment of malaria. There is limited evidence that artemether + lumefantrine and dihydroartemisinin + piperaquine achieve lower plasma concentrations in infants than in older children and adults.

Other considerations
The Guideline Development Group considered the currently available evidence too limited to warrant formal evidence review at this stage, and was unable to recommend any changes beyond the status quo. Further research is warranted.

The pharmacokinetics properties of many medicines in infants differ markedly from those in adults because of the physiological changes that occur in the first year of life (Annex 5). Accurate dosing is particularly important for infants. The only antimalarial agent that is currently contraindicated for infants (<6 months) is primaquine.

ACT is recommended and should be given according to body weight at the same mg/kg bw dose for all infants, including those weighing < 5 kg, with close monitoring of treatment response. The lack of infant formulations of most antimalarial drugs often necessitates division of adult tablets, which can lead to inaccurate dosing. When available, paediatric formulations and strengths are preferred, as they improve the effectiveness and accuracy of ACT dosing.

5.2.3 | OPTIMAL ANTIMALARIAL DOSING IN MALNOURISHED YOUNG CHILDREN

Malaria and malnutrition frequently coexist. Malnutrition may result in inaccurate dosing when doses are based on age (a dose may be too high for an infant with a low weight for age) or on weight (a dose may be too low for an infant with a low weight for age). Although many studies of the efficacy of antimalarial drugs have been conducted in populations and settings where malnutrition was prevalent, there are few studies of the disposition of the drugs specifically in malnourished individuals, and these seldom distinguished between acute and chronic malnutrition. Oral absorption of drugs may be reduced if there is diarrhoea or vomiting, or rapid gut transit or atrophy of the small bowel mucosa. Absorption of intramuscular and possibly intrarectal drugs may be slower, and diminished muscle mass may make it difficult to administer repeated intramuscular injections to malnourished patients. The volume of distribution of some drugs may be larger and the plasma concentrations lower. Hypoalbuminaemia may reduce protein binding and increase metabolic clearance, but concomitant hepatic dysfunction may reduce the metabolism of some drugs; the net result is uncertain.

Small studies of the pharmacokinetics of quinine and chloroquine showed alterations in people with different degrees of malnutrition. Studies of SP in IPTp and of amodiaquine monotherapy and dihydroartemisinin + piperaquine for treatment suggest reduced efficacy in malnourished children. A pooled analysis of data for individual patients showed that the concentrations of lumefantrine on day 7 were lower in children < 3 years who were underweight for age than in adequately nourished children and adults. Although these findings are concerning, they are insufficient to warrant dose modifications (in mg/kg bw) of any antimalarial drug in patients with malnutrition, however, their response to treatment should be monitored more closely.

5.3 | LARGE AND OBESE ADULTS

Large adults are at risk for under-dosing when they are dosed by age or in standard pre-packaged adult weight-based treatments. In principle, dosing of large adults should be based on achieving the target mg/kg bw dose for each antimalarial regimen. The practical consequence is that two packs of an antimalarial drug might have to be opened to ensure adequate treatment. For obese patients, less drug is often distributed to fat than to other tissues; therefore, they should be *dosed on the basis of an estimate of lean body weight*, *ideal body weight*. Patients who are heavy but not obese require the same mg/kg bw doses as lighter patients.

In the past, maximum doses have been recommended, but there is no evidence or justification for this practice. As the evidence for an association between dose, pharmacokinetics and treatment outcome in overweight or large adults is limited, and alternative dosing options have not been assessed in treatment trials, it is recommended that this gap in knowledge be assessed urgently. In the absence of data, treatment providers should attempt to follow up the treatment outcomes of large adults whenever possible.

5.4 | PATIENTS CO-INFECTED WITH HIV

There is considerable geographical overlap between malaria and HIV infection, and many people are co-infected. Worsening HIV-related immunosuppression may lead to more severe manifestations of malaria. In HIV-infected pregnant women, the adverse effects of placental malaria on birth weight are increased. In areas of stable endemic malaria, HIV-infected patients who are partially immune to malaria may have more frequent, higher-density infections, while in areas of unstable transmission, HIV infection is associated with increased risks for severe malaria and malaria-related deaths. Limited information is available on how HIV infection modifies therapeutic responses to ACTs. Early studies suggested that increasing HIV-related immunosuppression was associated with decreased treatment response to antimalarial drugs. There is presently insufficient information to modify the general malaria treatment recommendations for patients with HIV/AIDS.

Therapeutic interactions must be taken into consideration (See Annex 5, 5.14). Studies on prophylaxis with trimethoprim + sulfamethoxazole in HIV-infected children and adults show significant protection against malaria, even in areas with high rates of antifolate resistance. In studies of drug interactions between antiretroviral medicines and ACTs, HIV-co-infected individuals on trimethoprim + sulfamethoxazole and antiretroviral treatment, particularly zidovudine-containing regimens, had high rates of neutropenia when artesunate + amodiaquine was used for malaria treatment. HIV-infected children had a seven- to eightfold increased risk for neutropenia 14 days after starting of artesunate + amodiaquine than HIV-uninfected children. Hepatotoxicity has been documented when efavirenz was given with artesunate + amodiaquine, which may be due to inhibition of CYP2C8-

mediated amodiaquine metabolism by efavirenz. Data on the safety of nevirapine-based regimens in people receiving amodiaquine + artesunate are lacking, but lower levels of amodiaquine and its metabolite desethylamodiaquine have been reported when they were given together with nevirapine.

More data are available on use of artemether + lumefantrine with antiretroviral treatment. A study in children with uncomplicated malaria in a high-transmission area of Africa showed a decreased risk for recurrent malaria after treatment with artemether + lumefantrine in children receiving lopinavir–ritonavir-based antiretroviral treatment as compared with non-nucleoside reverse transcriptase inhibitor-based antiretroviral treatment. Evaluation of pharmacokinetics in these children and in healthy volunteers showed significantly higher exposure to lumefantrine and lower exposure to dihydroartemisinin with lopinavir–ritonavir-based antiretroviral treatment, but no adverse consequences. Conversely, efavirenz-based antiretroviral treatment was associated with a two- to fourfold decrease in exposure to lumefantrine in healthy volunteers and malaria-infected adults and children, with increased rates of recurrent malaria after treatment. Close monitoring is required. Increasing artemether + lumefantrine dosing with efavirenz-based antiretroviral treatment has not yet been studied. Exposure to lumefantrine and other non-nucleoside reverse transcriptase inhibitor-based antiretroviral treatment, namely nevirapine and etravirine, did not show consistent changes that would require dose adjustment.

Studies of administration of quinine with lopinavir–ritonavir or ritonavir alone in healthy volunteers gave conflicting results. The combined data are insufficient to justify dose adjustment. Single-dose atovaquone – proguanil with efavirenz, lopinavir–ritonavir or atazanavir–ritonavir were all associated with a significantly decreased area under the concentration–time curve for atovaquone (two- to fourfold) and proguanil (twofold), which could well compromise treatment or prophylactic efficacy. There is insufficient evidence to change the current mg/kg bw dosing recommendations; however, these patients should also be monitored closely.

5.5 | PATIENTS CO-INFECTED WITH TUBERCULOSIS

Rifamycins, in particular rifampicin, are potent CYP3A4 inducers with weak antimalarial activity. Concomitant administration of rifampicin during quinine treatment of adults with malaria was associated with a significant decrease in exposure to quinine and a five-fold higher recrudescence rate. Similarly, concomitant rifampicin with mefloquine in healthy adults was associated with a there-fold decrease in exposure to mefloquine. In adults co-infected with HIV and tuberculosis who were being treated with rifampicin, administration of artemether + lumefantrine resulted in significantly lower exposure to artemether, dihydroartemisinin and lumefantrine (nine-, six- and there-fold decreases, respectively). There is insufficient evidence at this time to change the current mg/kg bw dosing recommendations; however, as these patients are at higher risk of recrudescent infections they should be monitored closely.

5.6 | NON-IMMUNE TRAVELLERS

Non-immune travellers

Treat travellers with uncomplicated *P. falciparum* malaria returning to non-endemic settings with an ACT.
Strong recommendation, high-quality evidence

GRADE (see Annex 4, A4.1 and A4.2)
Studies have consistently demonstrated that the five WHO recommended ACTs have less than 5% PCR-adjusted treatment failure rates in settings without resistance to the partner drug (*high quality evidence*).

Other considerations
The Guideline Development Group considered the evidence of superiority of ACTs over non-ACTs from endemic settings to be equally applicable to those travelling from non-endemic settings.

Travellers who acquire malaria are often non-immune people living in cities in endemic countries with little or no transmission or are visitors from non-endemic countries travelling to areas with malaria transmission. Both are at higher risk for severe malaria. In a malaria-endemic country, they should be treated according to national policy, provided the treatment recommended has a recent proven cure rate > 90%. Travellers who return to a non-endemic country and then develop malaria present a particular problem, and the case fatality rate is often high; doctors in non-malarious areas may be unfamiliar with malaria and the diagnosis is commonly delayed, and effective antimalarial drugs may not be registered or may be unavailable. However prevention of transmission or the emergence of resistance are not relevant outside malaria-endemic areas. If the patient has taken chemoprophylaxis, the same medicine should not be used for treatment. Treatment of *P. vivax*, *P. ovale* and *P. malariae* malaria in travellers should be the same as for patients in endemic areas (see section 6).

There may be delays in obtaining artesunate, artemether or quinine for the management of severe malaria outside endemic areas. If only parenteral quinidine is available, it should be given, with careful clinical and electrocardiographic monitoring (see section 7).

5.7 | UNCOMPLICATED HYPERPARASITAEMIA

Uncomplicated hyperparasitaemia is present in patients who have ⩾ 4% parasitaemia but no signs of severity. They are at increased risk for severe malaria and for treatment failure and are considered an important source of antimalarial drug resistance. In falciparum malaria, the risk for progression to severe malaria with vital organ dysfunction increases at higher parasite densities. In low-transmission settings, mortality begins to increase when the parasite density exceeds 100 000/µL (~2% parasitaemia). On the north-west border of Thailand, before the general introduction of ACT, parasitaemia > 4% without signs of severity was associated with a 3% mortality rate (about 30-times higher than from uncomplicated falciparum malaria with lower densities) and a six-times higher risk of treatment failure. The relationship between parasitaemia and risks depends on the epidemiological context: in higher-transmission settings, the risk of developing severe malaria in patients with high parasitaemia is lower, but "uncomplicated hyperparasitaemia" is still associated with a significantly higher rate of treatment failure.

Patients with a parasitaemia of 4–10% and no signs of severity also require close monitoring, and, if feasible, admission to hospital. They have high rates of treatment failure. Non-immune people such as travellers and individuals in low-transmission settings with a parasitaemia > 2% are at increased risk and also require close attention. Parasitaemia > 10% is considered to indicate severe malaria in all settings.

It is difficult to make a general recommendation about treatment of uncomplicated hyperparasitaemia, for several reasons: recognizing these patients requires an accurate, quantitative parasite count (they will not be identified from semi-quantitative thick film counts or RDTs), the risks for severe malaria vary considerably, and the risks for treatment failure also vary. Furthermore, little information is available on therapeutic responses in uncomplicated hyperparasitaemia. As the artemisinin component of an ACT is essential in preventing progression to severe malaria, absorption of the first dose must be ensured (atovaquone – proguanil alone should not be used for travellers presenting with uncomplicated hyperparasitaemia). Longer courses of treatment are more effective; both giving longer courses of ACT and preceding the standard 3-day ACT regimen with parenteral or oral artesunate have been used.

3RD EDITION

6 | TREATMENT OF UNCOMPLICATED MALARIA CAUSED BY *P. VIVAX*, *P. OVALE*, *P. MALARIAE* OR *P. KNOWLESI*

Treating uncomplicated *P. vivax*, *P. ovale*, *P. malariae* or *P. knowlesi* malaria

Blood stage infection

If the malaria species is not known with certainty, treat as for uncomplicated *P. falciparum* malaria.
Good practice statement

In areas with chloroquine-susceptible infections, treat adults and children with uncomplicated *P. vivax*, *P. ovale*, *P. malariae* or *P. knowlesi* malaria with either an ACT (except pregnant women in their first trimester) or chloroquine.
Strong recommendation, high-quality evidence

In areas with chloroquine-resistant infections, treat adults and children with uncomplicated *P. vivax*, *P. ovale*, *P. malariae* or *P. knowlesi* malaria (except pregnant women in their first trimester) with an ACT.
Strong recommendation, high-quality evidence

Treat pregnant women in their first trimester who have chloroquine-resistant *P. vivax* malaria with quinine.
Strong recommendation, very low-quality evidence

Preventing relapse in *P. vivax* or *P. ovale* malaria

The G6PD status of patients should be used to guide administration of primaquine for preventing relapse.
Good practice statement

To prevent relapse, treat *P. vivax* or *P. ovale* malaria in children and adults (except pregnant women, infants aged < 6 months, women breastfeeding infants aged < 6 months, women breastfeeding older infants unless they are known no to be G6PD deficient, and people with G6PD deficiency) with a 14-day course (0.25-0.5 mg/kg bw daily) of primaquine in all transmission settings.
Strong recommendation, high-quality evidence

In people with G6PD deficiency, consider preventing relapse by giving primaquine base at 0.75 mg/kg bw once a week for 8 weeks, with close medical supervision for potential primaquine-induced adverse haematological effects.
Conditional recommendation, very low-quality evidence

When the G6PD status is unknown and G6PD testing is not available, a decision to prescribe primaquine must be based on an assessment of the risks and benefits of adding primaquine.
Good practice statement

Pregnant and breast feeding women

In women who are pregnant or breastfeeding, consider weekly chemoprophylaxis with chloroquine until delivery and breastfeeding are completed, then, on the basis of G6PD status, treat with primaquine to prevent future relapse.
Conditional recommendation, moderate-quality evidence

P. vivax is the second most important causative agent of human malaria. Approximately 35% of the world's population is at risk. *P. vivax* accounts for approximately 9%[8] of malaria cases worldwide and is the dominant malaria species outside Africa. *P. vivax* is prevalent in endemic areas in Asia, Central and South America, the Middle East and Oceania. In Africa, *P. vivax* is relatively uncommon, except in the Horn of Africa. In West Africa, *P. vivax* is rare except in Mauritania and Mali. In most areas in which *P. vivax* is prevalent, malaria transmission rates are low, and the affected populations therefore achieve little immunity. Consequently, people of all ages are at risk. The exception is the island of New Guinea, where transmission in some parts is intense. The other human malaria parasite species *P. malariae* and *P. ovale* (two sympatric species) are generally less prevalent, but they are distributed worldwide, especially in the tropical areas of Africa. During the past decade, there have been many reported human infections with *P. knowlesi,* a simian parasite in the forested areas of South-East Asia. In parts of the island of Borneo, this is now the main species causing malaria. Further information is provided in Annex 6.

Of the species of *Plasmodium* that affect humans, only *P. vivax* and *P. ovale* form hypnozoites, which are dormant parasite stages in the liver that cause relapses of infection weeks to years after the primary infection. Thus, a single mosquito inoculation may result in repeated bouts of illness. *P. vivax* exists in two general forms: the more prevalent tropical form, which causes malaria that relapses at frequent intervals (typically every 3 weeks unless slowly eliminated antimalarial drugs are given, in which case the interval is 5–7 weeks) and tends to be less susceptible to primaquine; and a temperate form, in which there may be a long (~9-month) incubation period or a similarly long interval between primary illness and relapse. The temperate form of *P. vivax* is more sensitive to primaquine. Infection with *P. vivax* during pregnancy reduces the birth weight of the infant, as does *P. falciparum*. In primigravidae, the birth weight reduction is approximately two thirds of that associated with *P. falciparum* (110 g compared with 170 g), but this adverse effect does not decrease with successive pregnancies, unlike in *P. falciparum* infections. *P. knowlesi* infections in humans are potentially dangerous; the patient may deteriorate rapidly because this parasite has a 24-h asexual cycle (quotidian) and so the parasite burden may expand rapidly, resulting in severe and sometimes fatal illness.

6.1 | THERAPEUTIC OBJECTIVE

The objective of treating malaria caused by *P. vivax* and *P. ovale* is to cure both blood-stage and liver-stage infections (called radical cure), thereby preventing recrudescence and relapse, respectively.

[8] WHO. World malaria report 2014. Geneva: World Health Organization; 2014 (http://who.int/malaria/publications/world_malaria_report_2014/en/).

6.2 | DIAGNOSIS

The clinical features of uncomplicated malaria are non-specific, and diagnosis of malaria requires blood testing. Malaria species are usually differentiated by microscopy. Young ring forms of all species look similar, but older stages and gametocytes have species-specific characteristics, except for the two forms of *P. ovale*, which appear identical. *P. knowlesi* malaria is frequently misdiagnosed as *P. malariae*, so any case of high parasitaemia with "*P. malariae-like*" parasites in or near an area where long- or pig-tailed macaque monkeys live should be treated as *P. knowlesi* until proved otherwise. *P. knowlesi* infections require confirmation by PCR. RDTs based on lateral flow immunochromatography are available for the detection of *P. vivax*, and their performance has improved in recent years, although their sensitivity for the other non-falciparum malarias is low. Molecular genotyping of *P. vivax* parasites is less useful in studies of therapeutic efficacy than in falciparum malaria as relapses may be due to either the same genotype that caused the initial illness or a different one.

6.3 | SUSCEPTIBILITY OF *P. VIVAX*, *P. OVALE*, *P. MALARIAE* AND *P. KNOWLESI* TO ANTIMALARIAL DRUGS

Few recent data are available on the susceptibility of *P. ovale, P. malariae* and *P. knowlesi* to antimalarial agents in vivo. These species are all regarded as sensitive to chloroquine, although chloroquine resistance was reported recently in *P. malariae*. Experience indicates that *P. ovale* and *P. malariae* are also susceptible to amodiaquine, mefloquine and the artemisinin derivatives and to ACT. Their susceptibility to antifolate antimalarial drugs, such as SP, is less certain. *P. knowlesi* is also sensitive to quinine, mefloquine, atovaquone – proguanil and artemether + lumefantrine and severe knowlesi malaria responds well to artesunate.

The susceptibility of *P. vivax* has been studied extensively, and, now that short-term culture methods have been standardized, the results of clinical studies are supported by *in vitro* observations. *P. vivax* is generally still sensitive to chloroquine, but resistance is increasing. High-level resistance to chloroquine is prevalent throughout the island of New Guinea, in Oceania and in parts of Indonesia. Lower-level resistance is found in other parts of South-East Asia and parts of South America. On the Indian subcontinent where most of the world's *P. vivax* malaria occurs, the parasites are mainly sensitive to chloroquine. Resistance to pyrimethamine has increased rapidly in some areas, rendering SP ineffective. There are insufficient data on current susceptibility to proguanil, although resistance to proguanil was selected rapidly when it was first used in areas endemic for *P. vivax* malaria.

In general, *P. vivax* is sensitive to all the other antimalarial drugs. In contrast to *P. falciparum*, asexual stages of *P. vivax* are also susceptible to primaquine. Thus, chloroquine + primaquine can be considered as a combination treatment for blood-stage infections, in addition to providing radical cure. The only drugs with significant activity against the hypnozoites are the 8-aminoquinolines (primaquine, bulaquine, tafenoquine). There is no standardized *in vitro* method for assessing the hypnozoiticidal activity of antimalarial drugs. *In vivo* assessments suggests that tolerance of *P. vivax* to primaquine is greater in eastern Asia and Oceania than elsewhere.

6.4 | TREATMENT OF BLOOD-STAGE INFECTION

Treating uncomplicated *P. vivax*, *P. ovale*, *P. malariae* or *P. knowlesi* malaria

In areas with chloroquine-susceptible infections, treat adults and children with uncomplicated *P. vivax*, *P. ovale, P. malariae* or *P. knowlesi* malaria with either an ACT (except pregnant women in their first trimester) or chloroquine.
Strong recommendation, high-quality evidence

In areas with chloroquine-resistant infections, treat adults and children with uncomplicated *P. vivax*, *P. ovale*, *P. malariae* or *P. knowlesi* malaria (except pregnant women in their first trimester) with an ACT.
Strong recommendation, high-quality evidence

GRADE (see Annex 4, A4.7 and A4.8)

In a systematic review of ACTs for the treatment of *P. vivax* malaria, five trials were conducted in Afghanistan, Cambodia, India, Indonesia and Thailand between 2002 and 2011 with a total of 1622 participants which compared ACTs directly with chloroquine. In comparison with chloroquine:

- ACTs cleared parasites from the peripheral blood more quickly (parasitaemia after 24 h of treatment: RR, 0.42; 95% CI, 0.36–0.50, four trials, 1652 participants, *high-quality evidence*); and
- ACTs were at least as effective in preventing recurrent parasitaemia before day 28 (RR, 0.58; 95% CI, 0.18–1.90, five trials, 1622 participants, *high-quality evidence*).

In four of these trials, few cases of recurrent parasitaemia were seen before day 28 with both chloroquine and ACTs. In the fifth trial, in Thailand in 2011, increased recurrent parasitaemia was seen after treatment with chloroquine (9%), but was infrequent after ACT (2%) (RR, 0.25; 95% CI, 0.09–0.66, one trial, 437 participants).

ACT combinations with long half-lives provided a longer prophylactic effect after treatment, with significantly fewer cases of recurrent parasitaemia between day 28 and day 42 or day 63 (RR, 0.57; 95% CI, 0.40–0.82, three trials, 1066 participants, *moderate-quality evidence*).

Other considerations

The guideline development group recognized that, in the few settings in which *P. vivax* is the only endemic species and where chloroquine resistance remains low, the increased cost of ACT may not be worth the small additional benefits. Countries where chloroquine is used for treatment of vivax malaria should monitor for chloroquine resistance and change to ACT when the treatment failure rate is > 10% at day 28.

Gogtay N, Kannan S, Thatte UM, Olliaro PL, Sinclair D. Artemisinin-based combination therapy for treating uncomplicated Plasmodium vivax malaria. Cochrane Database Systemat Rev 2013; 10:CD008492. doi: 10.1002/14651858.CD008492.pub3.

6.4.1 | UNCOMPLICATED *P. VIVAX* MALARIA

In areas with chloroquine-sensitive P. vivax

For chloroquine-sensitive vivax malaria, oral chloroquine at a total dose of 25 mg base/kg bw is effective and well tolerated. Lower total doses are not recommended, as these encourage the emergence of resistance. Chloroquine is given at an initial dose of 10 mg base/kg bw, followed by 10 mg/kg bw on the second day and 5 mg/kg bw on the third day. In the past, the initial 10-mg/kg bw dose was followed by 5 mg/kg bw at 6 h, 24 h and 48 h. As residual chloroquine suppresses the first relapse of tropical *P. vivax* (which emerges about 3 weeks after onset of the primary illness), relapses begin to occur 5–7 weeks after treatment if radical curative treatment with primaquine is not given.

ACTs are highly effective in the treatment of vivax malaria, allowing simplification (unification) of malaria treatment; i.e. all malaria infections can be treated with an ACT. The exception is artesunate + SP, where resistance significantly compromises its efficacy. Although good efficacy of artesunate + SP was reported in one study in Afghanistan, in several other areas (such as South-East Asia) *P. vivax* has become resistant to SP more rapidly than *P. falciparum*. The initial response to all ACTs is rapid in vivax malaria, reflecting the high sensitivity to artemisinin derivatives, but, unless primaquine is given, relapses commonly follow. The subsequent recurrence patterns differ, reflecting the elimination kinetics of the partner drugs. Thus, recurrences, presumed to be relapses, occur earlier after artemether + lumefantrine than after dihydroartemisinin + piperaquine or artesunate + mefloquine because lumefantrine is eliminated more rapidly than either mefloquine or piperaquine. A similar temporal pattern of recurrence with each of the drugs is seen in the *P. vivax* infections that follow up to one third of acute falciparum malaria infections in South-East Asia.

In areas with chloroquine-resistant P. vivax

ACTs containing piperaquine, mefloquine or lumefantrine are the recommended treatment, although artesunate + amodiaquine may also be effective in some areas.

In the systematic review of ACTs for treating *P. vivax* malaria, dihydroartemisinin + piperaquine provided a longer prophylactic effect than ACTs with shorter half-lives (artemether + lumefantrine, artesunate + amodiaquine), with significantly fewer recurrent parasitaemias during 9 weeks of follow-up (RR, 0.57; 95% CI, 0.40–0.82, three trials, 1066 participants). The half-life of mefloquine is similar to that of piperaquine, but use of dihydroartemisinin + piperaquine in *P. vivax* mono-infections has not been compared directly in trials with use of artesunate + mefloquine.

In the first-trimester of pregnancy, quinine should be used in place of ACTs (section 5.1).

6.4.2 | UNCOMPLICATED *P. OVALE*, *P. MALARIAE* OR *P. KNOWLESI* MALARIA

Resistance of *P. ovale, P. malariae* and *P. knowlesi* to antimalarial drugs is not well characterized, and infections caused by these three species are generally considered to be sensitive to chloroquine. In only one study, conducted in Indonesia, was resistance to chloroquine reported in *P. malariae*.

The blood stages of *P. ovale*, *P. malariae* and *P. knowlesi* should therefore be treated with the standard regimen of ACT or chloroquine, as for vivax malaria.

6.4.3 | MIXED MALARIA INFECTIONS

Mixed malaria infections are common in endemic areas. For example, in Thailand, despite low levels of malaria transmission, 8% of patients with acute vivax malaria also have *P. falciparum* infections, and one third of acute *P. falciparum* infections are followed by a presumed relapse of vivax malaria (making vivax malaria the most common complication of falciparum malaria).

Mixed infections are best detected by nucleic acid-based amplification techniques, such as PCR; they may be underestimated with routine microscopy. Cryptic *P. falciparum* infections in vivax malaria can be revealed in approximately 75% of cases by RDTs based on the *Pf*HRP2 antigen, but several RDTs cannot detect mixed infection or have low sensitivity for detecting cryptic vivax malaria. ACTs are effective against all malaria species and so are the treatment of choice for mixed infections.

6.5 | TREATMENT OF THE LIVER STAGES (HYPNOZOITES) OF *P. VIVAX* AND *P. OVALE*

Preventing relapse in *P. vivax* or *P. ovale* malaria

To prevent relapse, treat *P. vivax* or *P. ovale* malaria in children and adults (except pregnant women, infants aged < 6 months, women breastfeeding infants < 6 months, women breastfeeding older infants unless they are known not to be G6PD deficient and people with G6PD deficiency) with a 14-day course of primaquine in all transmission settings.
Strong recommendation, high-quality evidence

In people with G6PD deficiency, consider preventing relapse by giving primaquine base at 0.75 mg base/kg bw once a week for 8 weeks, with close medical supervision for potential primaquine-induced adverse haematological effects.
Conditional recommendation, very low-quality evidence

GRADE (see Annex 4, A4.9 and A4.10)
In a systematic review of primaquine for radical cure of *P. vivax* malaria, 14 days of primaquine was compared with placebo or no treatment in 10 trials, and 14 days was compared with 7 days in one trial. The trials were conducted in Colombia, Ethiopia, India, Pakistan and Thailand between 1992 and 2006.

In comparison with placebo or no primaquine:

- 14 days of primaquine (0.25 mg/kg bw per day) reduced relapses during 15 months of follow-up by about 40% (RR, 0.60; 95% CI, 0.48–0.75, 10 trials, 1740 participants, *high-quality evidence*).

In comparison with 7 days of primaquine:

- 14 days of primaquine (0.25 mg/kg bw per day) reduced relapses during 6 months of follow-up by over 50% (RR, 0.45; 95% CI, 0.25–0.81, one trial, 126 participants, *low-quality evidence*).

No direct comparison has been made of higher doses (0.5 mg/kg bw for 14 days) with the standard regimen (0.25 mg/kg bw for 14 days).

Twelve of the 15 trials included in the review explicitly excluded people with G6PD deficiency; the remaining three did not report on this aspect. No serious adverse events were reported.

Other considerations
In the absence of evidence to recommend alternatives, the guideline development group considers 0.75 mg/kg bw primaquine given once weekly for 8 weeks to be the safest regimen for people with mild-to-moderate G6PD deficiency.

Galappaththy GNL, Tharyan P, Kirubakaran R. Primaquine for preventing relapse in people with Plasmodium vivax malaria treated with chloroquine. Cochrane Database Systemat Rev 2013;10:CD004389. doi: 10.1002/14651858.CD004389.pub3.

6.5.1 | PRIMAQUINE FOR PREVENTING RELAPSE

To achieve radical cure (cure and prevention of relapse), relapses originating from liver hypnozoites must be prevented by giving primaquine. The frequency and pattern of relapses varies geographically, with relapse rates generally ranging from 8% to 80%. Temperate long-latency *P. vivax* strains are still prevalent in many areas. Recent evidence suggests that, in endemic areas where people are inoculated frequently with *P. vivax*, a significant proportion of the population harbours dormant but "activatable" hypnozoites. The exact mechanism of activation of dormant hypnozoites is unclear. There is evidence that systemic parasitic and bacterial infections, but not viral infections, can activate *P. vivax* hypnozoites, which explains why *P. vivax* commonly follows *P. falciparum* infections in endemic areas where both parasites are prevalent. Thus, the radical curative efficacy of primaquine must be set against the prevalent relapse frequency and the likely burden of "activatable" hypnozoites. Experimental studies on vivax malaria and the relapsing simian malaria *P. cynomolgi* suggest that the total dose of 8-aminoquinoline given is the main determinant of radical curative efficacy. In most therapeutic assessments, primaquine has been given for 14 days. Total doses of 3.5 mg base/kg bw (0.25 mg/kg bw per day) are required for temperate strains and 7 mg base/kg bw (0.5 mg/kg bw per day) is needed for the tropical, frequent-relapsing *P. vivax* prevalent in East Asia and Oceania. Primaquine causes dose-limiting abdominal discomfort when taken on an empty stomach; it should always be taken with food.

Primaquine formulation: If available, administer scored tablets containing 7.5 or 15 mg of primaquine. Smaller-dose tablets containing 2.5 and 5 mg base are available in some areas and facilitate accurate dosing in children. When scored tablets are not available, 5 mg tablets can be used.

Therapeutic dose: 0.25–0.5 mg/kg bw per day primaquine once a day for 14 days (see Annex 5, A5.11).

Use of primaquine to prevent relapse in high-transmission settings was not recommended previously, as the risk for new infections was considered to outweigh any benefits of preventing relapse. This may have been based on underestimates of the morbidity and mortality associated with multiple relapses, particularly in young children. Given the benefits of preventing relapse and in the light of changing epidemiology worldwide and more aggressive targets for malaria control and elimination, *the group now recommends that primaquine be used in all settings.*

6.5.2 | PRIMAQUINE AND GLUCOSE-6-PHOSPHATE DEHYDROGENASE DEFICIENCY

Any person (male or female) with red cell G6PD activity < 30% of the normal mean has G6PD deficiency and will experience haemolysis after primaquine. Heterozygote females with higher mean red cell activities may still show substantial haemolysis. G6PD deficiency is an inherited sex-linked genetic disorder, which is associated with some protection against *P. falciparum* and *P. vivax* malaria but increased susceptibility to oxidant haemolysis. The prevalence of G6PD deficiency varies, but in tropical areas it is typically 3–35%; high frequencies are found only in areas where malaria is or has been endemic. There are many (> 180) different G6PD deficiency genetic variants; nearly all of which make the red cells susceptible to oxidant haemolysis, but the severity of haemolysis may vary. Primaquine generates reactive intermediate metabolites that are oxidant and cause variable haemolysis in G6PD-deficient individuals. It also causes methemoglobinaemia. The severity of haemolytic anaemia depends on the dose of primaquine and on the variant of the G6PD enzyme. Fortunately, primaquine is eliminated rapidly so haemolysis is self-limiting once the drug is stopped. In the absence of exposure to primaquine or another oxidant agent, G6PD deficiency rarely causes clinical manifestationsso, many patients are unaware of their G6PD status. Screening for G6PD deficiency is not widely available outside hospitals, but rapid screening tests that can be used at points of care have recently become commercially available.

- In patients known to be G6PD deficient, primaquine may be considered at a dose of 0.75 mg base/kg bw once a week for 8 weeks. The decision to give or withhold primaquine should depend on the possibility of giving the treatment under close medical supervision, with ready access to health facilities with blood transfusion services.
- Some heterozygote females who test as normal or not deficient in qualitative G6PD screening tests have intermediate G6PD activity and can still haemolyse substantially. Intermediate deficiency (30–80% of normal) and normal enzyme activity (> 80% of normal) can be differentiated only with a quantitative test. In the absence of quantitative testing, all females should be considered as potentially having intermediate G6PD activity and given the 14-day regimen of primaquine, with counselling on how to recognize symptoms and signs of haemolytic anaemia. They should be advised to stop primaquine and be told where to seek care should these signs develop.
- If G6PD testing is not available, a decision to prescribe or withhold primaquine should be based on the balance of the probability and benefits of preventing relapse against the risks of primaquine-induced haemolytic anaemia. This depends on the population prevalence of G6PD deficiency, the severity of the prevalent genotypes and on the capacity of health services to identify and manage primaquine-induced haemolytic reactions.

6.5.3 | PREVENTION OF RELAPSE IN PREGNANT OR LACTATING WOMEN AND INFANTS

Preventing relapse in pregnant or lactating women

In women who are pregnant or breastfeeding, consider weekly chemoprophylaxis with chloroquine until delivery and breastfeeding are completed, then, on the basis of G6PD status, treat with primaquine to prevent future relapse.
Conditional recommendation, moderate-quality evidence

GRADE (see Annex 4, A4.11)
In a systematic review of malaria chemoprophylaxis in pregnant women, chloroquine prophylaxis against *P. vivax* during pregnancy was directly evaluated in one trial conducted in Thailand in 2001. In comparison with no chemoprophylaxis:

- Chloroquine prophylaxis substantially reduced recurrent *P. vivax* malaria (RR, 0.02; 95% CI, 0.00–0.26, one trial, 951 participants, moderate-quality evidence).

Radeva-Petrova D, ter Kuile FO, Sinclair D, Kayentao K, Garner P. Drugs for preventing malaria in pregnant women in endemic areas: any drug regimen versus placebo/no treatment. Cochrane Database Systemat Rev 2014; 10:CD000169. doi: 10.1002/14651858.CD000169.pub3.

Primaquine is contraindicated in pregnant women, infants < 6months of age and in lactating women (unless the infant is known not to be G6PD deficient).

As an alternative, chloroquine prophylaxis could be given to suppress relapses after acute vivax malaria during pregnancy. Once the infant has been delivered and the mother has completed breastfeeding, primaquine could then be given to achieve radical cure.

Few data are available on the safety of primaquine in infancy, and in the past primaquine was not recommended for infants. There is, however, no specific reason why primaquine should not be given to children aged 6 months to 1 year (provided they do not have G6PD deficiency), as this age group may suffer multiple relapses from vivax malaria. The guideline development group therefore recommended lowering the age restriction to 6 months.

3RD EDITION

7 | TREATMENT OF SEVERE MALARIA

Treating severe malaria

Treat adults and children with severe malaria (including infants, pregnant women in all trimesters and lactating women) with intravenous or intramuscular artesunate for at least 24 h and until they can tolerate oral medication. Once a patient has received at least 24 h of parenteral therapy and can tolerate oral therapy, complete treatment with 3 days of an ACT (add single dose primaquine in areas of low transmission).
Strong recommendation, high-quality evidence

Revised dose recommendation for parenteral artesunate in young children

Children weighing < 20 kg should receive a higher dose of artesunate (3 mg/kg bw per dose) than larger children and adults (2.4 mg/kg bw per dose) to ensure equivalent exposure to the drug.
Strong recommendation based on pharmacokinetic modelling

Parenteral alternatives when artesunate is not available

If parenteral artesunate is not available, use artemether in preference to quinine for treating children and adults with severe malaria.
Conditional recommendation, low-quality evidence

Treating cases of suspected severe malaria pending transfer to a higher-level facility (pre-referral treatment)

Pre-referral treatment options

Where complete treatment of severe malaria is not possible but injections are available, give adults and children a single intramuscular dose of artesunate, and refer to an appropriate facility for further care. Where intramuscular artesunate is not available use intramuscular artemether or, if that is not available, use intramuscular quinine.
Strong recommendation, moderate-quality evidence

Where intramuscular injections of artesunate are not available, treat children < 6 years with a single rectal dose (10 mg/kg bw) of artesunate, and refer immediately to an appropriate facility for further care. Do not use rectal artesunate in older children and adults.
Strong recommendation, moderate-quality evidence

Mortality from untreated severe malaria (particularly cerebral malaria) approaches 100%. With prompt, effective antimalarial treatment and supportive care, the rate falls to 10–20% overall. Within the broad definition of severe malaria some syndromes are associated with lower mortality rates (e.g. severe anaemia) and others with higher mortality rates (e.g. acidosis). The risk for death increases in the presence of multiple complications.

Any patient with malaria who is unable to take oral medications reliably, shows any evidence of vital organ dysfunction or has a high parasite count is at increased risk for dying. The exact risk depends on the species of infecting malaria parasite, the number of systems affected, the degree of vital organ dysfunction, age, background immunity, pre-morbid, and concomitant diseases, and access to appropriate treatment. Tests such as a parasite count, haematocrit and blood glucose may all be performed immediately at the point of care, but the results of other laboratory measures, if any, may be available only after hours or days. As severe malaria is potentially fatal, any patient considered to be at increased risk should be given the benefit of the highest level of care available. The attending clinician should not worry unduly about definitions: the severely ill patient requires immediate supportive care, and, if severe malaria is a possibility, parenteral antimalarial drug treatment should be started without delay.

7.1 | DEFINITIONS

7.1.1 | SEVERE FALCIPARUM MALARIA

For epidemiological purposes, **severe falciparum malaria** is defined as one or more of the following, occurring in the absence of an identified alternative cause and in the presence of *P. falciparum* asexual parasitaemia.

- *Impaired consciousness:* A Glasgow coma score < 11 in adults or a Blantyre coma score < 3 in children
- *Prostration:* Generalized weakness so that the person is unable to sit, stand or walk without assistance
- *Multiple convulsions:* More than two episodes within 24 h
- *Acidosis:* A base deficit of > 8 mEq/L or, if not available, a plasma bicarbonate level of < 15 mmol/L or venous plasma lactate ⩾ 5 mmol/L. Severe acidosis manifests clinically as respiratory distress (rapid, deep, laboured breathing).
- *Hypoglycaemia:* Blood or plasma glucose < 2.2 mmol/L (< 40 mg/dL)
- *Severe malarial anaemia:* Haemoglobin concentration ⩽ 5 g/dL or a haematocrit of ⩽ 15% in children < 12 years of age (< 7 g/dL and < 20%, respectively, in adults) with a parasite count > 10 000/µL
- *Renal impairment:* Plasma or serum creatinine > 265 µmol/L (3 mg/dL) or blood urea > 20 mmol/L
- *Jaundice:* Plasma or serum bilirubin > 50 µmol/L (3 mg/dL) with a parasite count > 100 000/ µL
- *Pulmonary oedema*: Radiologically confirmed or oxygen saturation < 92% on room air with a respiratory rate > 30/min, often with chest indrawing and crepitations on auscultation
- *Significant bleeding:* Including recurrent or prolonged bleeding from the nose, gums or venepuncture sites; haematemesis or melaena

- *Shock:* Compensated shock is defined as capillary refill ⩾ 3 s or temperature gradient on leg (mid to proximal limb), but no hypotension. Decompensated shock is defined as systolic blood pressure < 70 mm Hg in children or < 80 mm Hg in adults, with evidence of impaired perfusion (cool peripheries or prolonged capillary refill).
- *Hyperparasitaemia: P. falciparum* parasitaemia > 10%.

7.1.2 | SEVERE VIVAX AND KNOWLESI MALARIA

Severe vivax malaria is defined as for falciparum malaria but with no parasite density thresholds.

Severe knowlesi malaria is defined as for falciparum malaria but with two differences:

- *P. knowlesi* hyperparasitaemia: parasite density > 100 000/μL
- Jaundice and parasite density > 20 000/μL.

7.2 | THERAPEUTIC OBJECTIVES

The main objective of the treatment of severe malaria is to prevent the patient from dying. Secondary objectives are prevention of disabilities and prevention of recrudescent infection.

Death from severe malaria often occurs within hours of admission to a hospital or clinic, so it is essential that therapeutic concentrations of a highly effective antimalarial drug be achieved as soon as possible. Management of severe malaria comprises mainly clinical assessment of the patient, specific antimalarial treatment, additional treatment and supportive care.

7.3 | CLINICAL ASSESSMENT

Severe malaria is a medical emergency. An open airway should be secured in unconscious patients and breathing and circulation assessed. The patient should be weighed or body weight estimated, so that medicines, including antimalarial drugs and fluids, can be given appropriately. An intravenous cannula should be inserted, and blood glucose (rapid test), haematocrit or haemoglobin, parasitaemia and, in adults, renal function should be measured immediately. A detailed clinical examination should be conducted, including a record of the coma score. Several coma scores have been advocated: the Glasgow coma scale is suitable for adults, and the simple Blantyre modification is easily performed in children. Unconscious patients should undergo a lumbar puncture for cerebrospinal fluid analysis to exclude bacterial meningitis.

The degree of acidosis is an important determinant of outcome; the plasma bicarbonate or venous lactate concentration should be measured, if possible. If facilities are available, arterial or capillary blood pH and gases should be measured in patients who are unconscious, hyperventilating or in shock. Blood should be taken for cross-matching, a full blood count, a platelet count, clotting studies, blood culture and full biochemistry (if possible). Careful attention should be paid to the patient's fluid balance in severe malaria in order to avoid over- or under-hydration. Individual requirements vary widely and depend on fluid losses before admission.

The differential diagnosis of fever in a severely ill patient is broad. Coma and fever may be due to meningoencephalitis or malaria. Cerebral malaria is not associated with signs of meningeal irritation (neck stiffness, photophobia or Kernig's sign), but the patient may be opisthotonic. As untreated bacterial meningitis is almost invariably fatal, a diagnostic lumbar puncture should be performed to exclude this condition. There is also considerable clinical overlap between septicaemia, pneumonia and severe malaria, and these conditions may coexist. When possible, blood should always be taken on admission for bacterial culture. In malaria-endemic areas, particularly where parasitaemia is common in young age groups, it is difficult to rule out septicaemia immediately in a shocked or severely ill obtunded child. *In all such cases, empirical parenteral broad-spectrum antibiotics should be started immediately, together with antimalarial treatment.*

7.4 | TREATMENT OF SEVERE MALARIA

It is essential that full doses of effective parenteral (or rectal) antimalarial treatment be given promptly in the initial treatment of severe malaria. This should be followed by a full dose of effective ACT orally. Two classes of medicine are available for parenteral treatment of severe malaria: artemisinin derivatives (artesunate or artemether) and the cinchona alkaloids (quinine and quinidine). Parenteral artesunate is the treatment of choice for all severe malaria. The largest randomized clinical trials ever conducted on severe falciparum malaria showed a substantial reduction in mortality with intravenous or intramuscular artesunate as compared with parenteral quinine. The reduction in mortality was not associated with an increase in neurological sequelae in artesunate-treated survivors. Furthermore, artesunate is simpler and safer to use.

7.4.1 | ARTESUNATE

Treating severe malaria

Treat adults and children with severe malaria (including infants, pregnant women in all trimesters and lactating women) with intravenous or intramuscular artesunate for at least 24 h. Once a patient has received at least 24 h of parenteral therapy and can tolerate oral therapy, complete treatment with 3 days of an ACT.
Strong recommendation, high-quality evidence

GRADE (see Annex 4, A4.12)
In a systematic review of artesunate for severe malaria, eight randomized controlled trials with a total of 1664 adults and 5765 children, directly compared parenteral artesunate with parenteral quinine. The trials were conducted in various African and Asian countries between 1989 and 2010.
In comparison with quinine, parenteral artesunate:

- Reduced mortality from severe malaria by about 40% in adults (RR, 0.61; 95% CI, 0.50–0.75, five trials, 1664 participants, *high-quality evidence*);
- Reduced mortality from severe malaria by about 25% in children (RR, 0.76; 95% CI, 0.65–0.90, four trials, 5765 participants, *high-quality evidence*); and
- Was associated with a small increase in neurological sequelae in children at the time of hospital discharge (RR, 1.36; 95% CI, 1.01–1.83, three trials, 5163 participants, *moderate-quality evidence*), most of which, however, slowly resolved, with little or no difference between artesunate and quinine 28 days later (*moderate-quality evidence*).

Other considerations
The guideline development group considered that the small increase in neurological sequelae at discharge after treatment with artesunate was due to the delayed recovery of the severely ill patients, who would have died had they received quinine. This should not be interpreted as a sign of neurotoxicity.
Although the safety of artesunate given in the first trimester of pregnancy has not been firmly established, the guideline development group considered that the proven benefits to the mother outweigh any potential harm to the developing fetus.

Sinclair D, Donegan S, Isba R, Lalloo DG. Artesunate versus quinine for treating severe malaria. Cochrane Database Systemat Rev 2012;6:CD005967. doi: 10.1002/14651858.CD005967.pub4.

Dosing of artesunate injection in severe malaria

Revised dose recommendation for parenteral artesunate in young children with severe malaria

Children weighing less than 20 kg should receive a higher parenteral dose of artesunate (3 mg/kg/dose) than larger children and adults (2.4 mg/kg/dose) to ensure equivalent drug exposure.
Strong recommendation based on pharmacokinetic modelling

The dosing subgroup reviewed all available pharmacokinetic data on artesunate and the main biologically active metabolite dihydroartemisinin following administration of artesunate in severe malaria (published pharmacokinetic studies from 71 adults and 265 children). Simulations of artesunate and dihydroartemisinin exposures were conducted for each age group. These showed underexposure in younger children. The revised parenteral dose regimens are predicted to provide equivalent artesunate and dihydroartemisinin exposures across all age groups.

Other considerations
Individual parenteral artesunate doses between 1.75 and 4 mg/kg have been studied and no toxicity has been observed. The GRC concluded that the predicted benefits of improved antimalarial exposure in children are not at the expense of increased risk.

Hendriksen IC, Mtove G, Kent A, Gesase S, Reyburn H, Lemnge MM, et al. Population pharmacokinetics of intramuscular artesunate in African children with severe malaria: implications for a practical dosing regimen. Clin Pharmacol Ther 2013;93:443–50.

Zaloumis SG, Tarning J, Krishna S, Price RN, White NJ, Davis TM, McCaw JM, Olliaro P, Maude RJ, Kremsner P, Dondorp A, Gomes M, Barnes K, Simpson JA. Population pharmacokinetics of intravenous artesunate: a pooled analysis of individual data from patients with severe malaria. CPT Pharmacometrics Syst Pharmacol. 2014;3:e145.

Artesunate is dispensed as a powder of artesunic acid, which is dissolved in sodium bicarbonate (5%) to form sodium artesunate. The solution is then diluted in approximately 5 mL of 5% dextrose and given by intravenous injection or by intramuscular injection into the anterior thigh.

The solution should be prepared freshly for each administration and should not be stored. Artesunate is rapidly hydrolysed in-vivo to dihydroartemisinin, which provides the main antimalarial effect. Studies of the pharmacokinetics of parenteral artesunate in children with severe malaria suggest that they have less exposure than older children and adults to both artesunate and the biologically active metabolite dihydroartemisinin. Body weight has been identified as a significant covariate in studies of the pharmacokinetics of orally and rectally administered artesunate, which suggests that young children have a larger apparent volume of distribution for both compounds and should therefore receive a slightly higher dose of parenteral artesunate to achieve exposure comparable to that of older children and adults.

Artesunate and post-treatment haemolysis

Delayed haemolysis starting >1 week after artesunate treatment of severe malaria has been reported in hyperparasitaemic non-immune travellers. Between 2010 and 2012, there were six reports involving a total of 19 European travellers with severe malaria who were treated with artesunate injection and developed delayed haemolysis. All except one were adults (median age, 50 years; range, 5–71 years). In a prospective study involving African children, the same phenomenon was reported in 5 (7%) of the 72 hyperparasitaemic children studied. Artesunate rapidly kills ring-stage parasites, which are then taken out of the red cells by the spleen; these infected erythrocytes are then returned to the circulation but with a shortened life span, resulting in the observed haemolysis. Thus, post-treatment haemolysis is a predictable event related to the life-saving effect of artesunate. Hyperparasitaemic patients must be followed up carefully to identify late-onset anaemia.

7.4.2 | PARENTERAL ALTERNATIVES WHEN ARTESUNATE IS NOT AVAILABLE

Parenteral alternatives when artesunate is not available

If parenteral artesunate is not available, use intramuscular artemether in preference to quinine for treating children and adults with severe malaria.
Strong recommendation, high-quality evidence

GRADE (see Annex 4, A4.13 and A4.14)
A systematic review of intramuscular artemether for severe malaria comprised two randomized controlled trials in Viet Nam in which artemether was compared with artesunate in 494 adults, and 16 trials in Africa and Asia in which artemether was compared with quinine in 716 adults and 1447 children. The trials were conducted between 1991 and 2009.

In comparison with artesunate, intramuscular artemether was not as effective at preventing deaths in adults in Asia (RR, 1.80; 95% CI, 1.09–2.97; two trials, 494 participants, *moderate-quality evidence*).

Artemether and artesunate have not been directly compared in randomized trials in African children.

In comparison with quinine:

- Intramuscular artemether prevented a similar number of deaths in children in Africa (RR, 0.96; 95% CI, 0.76–1.20; 12 trials, 1447 participants, *moderate-quality evidence*).
- Intramuscular artemether prevented more deaths in adults in Asia (RR, 0.59; 95% CI, 0.42–0.83; four trials, 716 participants, *moderate-quality evidence*).

Other considerations
Indirect comparisons of parenteral artesunate and quinine and of artemether and quinine were considered by the guideline development group with what is known about the pharmacokinetics of the two drugs. They judged the accumulated indirect evidence to be sufficient to recommend parenteral artesunate rather than intramuscular artemether for use in all age groups.

Esu E, Effa EE, Opie ON, Uwaoma A, Meremikwu MM. Artemether for severe malaria. Cochrane Database Systemat Rev 2014;9:CD010678. doi: 10.1002/14651858.CD010678.pub2.

Artemether

Artemether is two to three times less active than its main metabolite dihydroartemisinin. Artemether can be given as an oil-based intramuscular injection or orally. In severe falciparum malaria, the concentration of the parent compound predominates after intramuscular injection, whereas parenteral artesunate is hydrolysed rapidly and almost completely to dihydroartemisinin. Given intramuscularly, artemether may be absorbed more slowly and more erratically than water-soluble artesunate, which is absorbed rapidly and reliably after intramuscular injection. These pharmacological advantages may explain the clinical superiority of parenteral artesunate over artemether in severe malaria.

Artemether is dispensed dissolved in oil (groundnut, sesame seed) and given by intramuscular injection into the anterior thigh.

Therapeutic dose: The initial dose of artemether is 3.2 mg/kg bw intramuscularly (to the anterior thigh). The maintenance dose is 1.6 mg/kg bw intramuscularly daily.

Quinine

Quinine treatment for severe malaria was established before the methods for modern clinical trials were developed. Several salts of quinine have been formulated for parenteral use, but the dihydrochloride is the most widely used. The peak concentrations after intramuscular quinine in severe malaria are similar to those after intravenous infusion. Studies of pharmacokinetics show that a loading dose of quinine (20 mg salt/kg bw, twice the maintenance dose) provides therapeutic plasma concentrations within 4 h. The maintenance dose of quinine (10 mg salt/kg bw) is administered at 8-h intervals, starting 8 h after the first dose. If there is no improvement in the patient's condition within 48 h, the dose should be reduced by one third, i.e. to 10 mg salt/kg bw every 12 h.

Rapid intravenous administration of quinine is dangerous. Each dose of parenteral quinine must be administered as a slow, rate-controlled infusion (usually diluted in 5% dextrose and infused over 4 h). The infusion rate should not exceed 5 mg salt/kg bw per h.

Whereas many antimalarial drugs are prescribed in terms of base, for historical reasons quinine doses are usually recommended in terms of salt (usually sulphate for oral use and dihydrochloride for parenteral use). Recommendations for the doses of this and other antimalarial agents should state clearly whether the salt or the base is being referred to; doses with different salts must have the same base equivalents. Quinine must never be given by intravenous bolus injection, as lethal hypotension may result.

Quinine dihydrochloride should be given by rate-controlled infusion in saline or dextrose solution. If this is not possible, it should be given by intramuscular injection to the anterior thigh; quinine should not be injected into the buttock in order to avoid sciatic nerve injury. The first dose should be split, with 10 mg/kg bw into each thigh. Undiluted quinine dihydrochloride at a concentration of 300 mg/mL is acidic (pH 2) and painful when given by intramuscular injection, so it is best to administer it either in a buffered formulation or diluted to a concentration of 60–100 mg/mL for intramuscular injection. Gluconate salts are less acidic and better tolerated than the dihydrochloride salt when given by the intramuscular and rectal routes.

As the first (loading) dose is the most important in the treatment of severe malaria, it should be reduced only if there is clear evidence of adequate pre-treatment before presentation. Although quinine can cause hypotension if administered rapidly, and overdose is associated with blindness and deafness, these adverse effects are rare in the treatment of severe malaria. The dangers of insufficient treatment (i.e. death from malaria) exceed those of excessive initial treatment.

7.5 | PRE-REFERRAL TREATMENT OPTIONS

Treating cases of suspected severe malaria pending transfer to higher-level facilities (pre-referral treatment)

Where complete treatment of severe malaria is not possible but injections are available, give adults and children a single intramuscular dose of artesunate, and refer to an appropriate facility for further care. Where intramuscular artesunate is not available use intramuscular artemether or, if that is not available, use intramuscular quinine.
Strong recommendation, moderate-quality evidence

Where intramuscular injections of artesunate are not available, treat children < 6 years with a single rectal dose (10mg/kg bw) of artesunate, and refer immediately to an appropriate facility for further care. Do not use rectal artesunate in older children and adults.
Strong recommendation, moderate-quality evidence

GRADE (see Annex 4, A4.15 and A4.16)
In a systematic review of pre-referral treatment for suspected severe malaria, in a single large randomized controlled trial of 17 826 children and adults in Bangladesh, Ghana and the United Republic of Tanzania, pre-referral rectal artesunate was compared with placebo.

In comparison with placebo:

- Rectal artesunate reduced mortality by about 25% in children < 6 years (RR, 0.74; 95% CI, 0.59–0.93; one trial, 8050 participants, *moderate-quality evidence*).
- Rectal artesunate was associated with more deaths in older children and adults (RR, 2.21; 95% CI, 1.18–4.15; one trial 4018 participants, *low-quality evidence*).

Other considerations
The guideline development group could find no plausible explanation for the finding of increased mortality among older children and adults in Asia who received rectal artesunate, which may be due to chance. Further trials would provide clarification but are unlikely to be done. The group was therefore unable to recommend its use in older children and adults.

In the absence of direct evaluations of parenteral antimalarial drugs for pre-referral treatment, the guideline development group considered the known benefits of artesunate in hospitalized patients and downgraded the quality of evidence for pre-referral situations. When intramuscular injections can be given, the group recommends intramuscular artesunate in preference to rectal artesunate.

Okebe J, Eisenhut M. Pre-referral rectal artesunate for severe malaria. Cochrane Database Systemat Rev 2014;5:CD009964. doi: 10.1002/14651858.CD009964.pub2.

The risk for death from severe malaria is greatest in the first 24 h, yet, in most malaria-endemic countries, the transit time between referral and arrival at a health facility where intravenous treatment can be administered is usually long, thus delaying the start of appropriate antimalarial treatment. During this time, the patient may deteriorate or die. It is therefore recommended that patients, particularly young children, be treated with a first dose of one of the recommended treatments before referral (unless the referral time is < 6 h).

The recommended pre-referral treatment options for children < 6 years, in descending order of preference, are intramuscular artesunate; rectal artesunate; intramuscular artemether; and intramuscular quinine. For older children and adults, the recommended pre-referral treatment options, in descending order of preference, are intramuscular injections of artesunate; artemether; and quinine.

Administration of an artemisinin derivative by the rectal route as pre-referral treatment is feasible and acceptable even at community level. The only trial of rectal artesunate as pre-referral treatment showed the expected reduction in mortality of young children but unexpectedly found increased mortality in older children and adults. As a consequence, rectal artesunate is recommended for use only in children aged < 6 years and only when intramuscular artesunate is not available.

When rectal artesunate is used, patients should be transported immediately to a higher-level facility where intramuscular or intravenous treatment is available. If referral is impossible, rectal treatment could be continued until the patient can tolerate oral medication. At this point, a full course of the recommended ACT for uncomplicated malaria should be administered.

The single dose of 10 mg/kg bw of artesunate when given as a suppository should be administered rectally as soon as a presumptive diagnosis of severe malaria is made. If the suppository is expelled from the rectum within 30 min of insertion, a second suppository should be inserted and the buttocks held together for 10 min to ensure retention of the dose.

7.6 | ADJUSTMENT OF PARENTERAL DOSING IN RENAL FAILURE OR HEPATIC DYSFUNCTION

The dosage of artemisinin derivatives does not have to be adjusted for patients with vital organ dysfunction. However quinine accumulates in severe vital organ dysfunction. If a patient with severe malaria has persisting acute kidney injury or there is no clinical improvement by 48 h, the dose of quinine should be reduced by one third, to 10 mg salt/kg bw every 12 h. Dosage adjustments are not necessary if patients are receiving either haemodialysis or haemofiltration.

7.7 | FOLLOW-ON TREATMENT

The current recommendation of experts is to give parenteral antimalarial drugs for the treatment of severe malaria for a minimum of 24 h once started (irrespective of the patient's ability to tolerate oral medication earlier) or until the patient can tolerate oral medication, before giving the oral follow-up treatment.

After initial parenteral treatment, once the patient can tolerate oral therapy, it is essential to continue and complete treatment with an effective oral antimalarial drug by giving a full course of effective ACT (artesunate + amodiaquine, artemether + lumefantrine or dihydroartemisinin + piperaquine). If the patient presented initially with impaired consciousness, ACTs containing mefloquine should be avoided because of an increased incidence of neuropsychiatric complications. When an ACT is not available, artesunate + clindamycin, artesunate + doxycycline, quinine + clindamycin or quinine + doxycycline can be used for follow-on treatment. Doxycycline is preferred to other tetracyclines because it can be given once daily and does not accumulate in cases of renal failure, but it should not be given to children < 8 years or pregnant women. As treatment with doxycycline is begun only when the patient has recovered sufficiently, the 7-day doxycycline course finishes after the artesunate, artemether or quinine course. When available, clindamycin is preferred in children and pregnant women.

7.8 | CONTINUING SUPPORTIVE CARE

Patients with severe malaria require intensive nursing care, preferably in an intensive care unit where possible. Clinical observations should be made as frequently as possible and should include monitoring of vital signs, coma score and urine output. Blood glucose should be monitored every 4 h, if possible, particularly in unconscious patients.

7.9 | MANAGEMENT OF COMPLICATIONS

Severe malaria is associated with a variety of manifestations and complications, which must be recognized promptly and treated as shown below.

Immediate clinical management of severe manifestations and complications of *P. falciparum* malaria

Manifestation or complication	Immediate management[a]
Coma (cerebral malaria)	Maintain airway, place patient on his or her side, exclude other treatable causes of coma (e.g. hypoglycaemia, bacterial meningitis); avoid harmful ancillary treatments, intubate if necessary.
Hyperpyrexia	Administer tepid sponging, fanning, a cooling blanket and paracetamol.
Convulsions	Maintain airways; treat promptly with intravenous or rectal diazepam, lorazepam, midazolam or intramuscular paraldehyde. Check blood glucose.
Hypoglycaemia	Check blood glucose, correct hypoglycaemia and maintain with glucose-containing infusion. Although hypoglycaemia is defined as glucose < 2.2 mmol/L, the threshold for intervention is < 3 mmol/L for children < 5 years and < 2.2 mmol/L for older children and adults.
Severe anaemia	Transfuse with screened fresh whole blood.
Acute pulmonary oedema[b]	Prop patient up at an angle of 45°, give oxygen, give a diuretic, stop intravenous fluids, intubate and add positive end-expiratory pressure or continuous positive airway pressure in life-threatening hypoxaemia.
Acute kidney injury	Exclude pre-renal causes, check fluid balance and urinary sodium; if in established renal failure, add haemofiltration or haemodialysis, or, if not available, peritoneal dialysis.
Spontaneous bleeding and coagulopathy	Transfuse with screened fresh whole blood (cryoprecipitate, fresh frozen plasma and platelets, if available); give vitamin K injection.

Metabolic acidosis	Exclude or treat hypoglycaemia, hypovolaemia and septicaemia. If severe, add haemofiltration or haemodialysis.
Shock	Suspect septicaemia, take blood for cultures; give parenteral broad-spectrum antimicrobials, correct haemodynamic disturbances.

[a] It is assumed that appropriate antimalarial treatment will have been started in all cases.

[b] Prevent by avoiding excess hydration

7.10 | ADDITIONAL ASPECTS OF MANAGEMENT

7.10.1 | FLUID THERAPY

Fluid requirements should be assessed individually. Adults with severe malaria are very vulnerable to fluid overload, while children are more likely to be dehydrated. The fluid regimen must also be adapted to the infusion of antimalarial drugs. Rapid bolus infusion of colloid or crystalloids is contraindicated. If available, haemofiltration should be started early for acute kidney injury or severe metabolic acidosis, which do not respond to rehydration. As the degree of fluid depletion varies considerably in patients with severe malaria, it is not possible to give general recommendations on fluid replacement; each patient must be assessed individually and fluid resuscitation based on the estimated deficit. In high-transmission settings, children commonly present with severe anaemia and hyperventilation (sometimes termed "respiratory distress") resulting from severe metabolic acidosis and anaemia; they should be treated by blood transfusion. In adults, there is a very thin dividing line between over-hydration, which may produce pulmonary oedema, and under-hydration, which contributes to shock, worsening acidosis and renal impairment. Careful, frequent evaluation of jugular venous pressure, peripheral perfusion, venous filling, skin turgor and urine output should be made.

7.10.2 | BLOOD TRANSFUSION

Severe malaria is associated with rapid development of anaemia, as infected, once infected and uninfected erythrocytes are haemolysed and/or removed from the circulation by the spleen. Ideally, fresh, cross-matched blood should be transfused; however, in most settings, cross-matched virus-free blood is in short supply. As for fluid resuscitation, there are not enough studies to make strong evidence-based recommendations on the indications for transfusion; the recommendations given here are based on expert opinion. In high-transmission settings, blood transfusion is generally recommended for children with a haemoglobin level of < 5 g/100 mL (haematocrit < 15%). In low-transmission settings, a threshold of 20% (haemoglobin, 7 g/100 mL) is recommended. These general recommendations must, however, be adapted to the individual, as the pathological consequences of rapid development of anaemia are worse than those of chronic or acute anaemia when there has been adaptation and a compensatory right shift in the oxygen dissociation curve.

7.10.3 | EXCHANGE BLOOD TRANSFUSION

Many anecdotal reports and several series have claimed the benefit of exchange blood transfusion in severe malaria, but there have been no comparative trials, and there is no consensus on whether it reduces mortality or how it might work. Various rationales have been proposed:

- removing infected red blood cells from the circulation and therefore lowering the parasite burden (although only the circulating, relatively non-pathogenic stages are removed, and this is also achieved rapidly with artemisinin derivatives);
- rapidly reducing both the antigen load and the burden of parasite-derived toxins, metabolites and toxic mediators produced by the host; and
- replacing the rigid unparasitized red cells by more easily deformable cells, therefore alleviating microcirculatory obstruction.

Exchange blood transfusion requires intensive nursing care and a relatively large volume of blood, and it carries significant risks. There is no consensus on the indications, benefits and dangers involved or on practical details such as the volume of blood that should be exchanged. It is, therefore, not possible to make any recommendation regarding the use of exchange blood transfusion.

7.10.4 | CONCOMITANT USE OF ANTIBIOTICS

The threshold for administering antibiotic treatment should be low in severe malaria. Septicaemia and severe malaria are associated, and there is substantial diagnostic overlap, particularly in children in areas of moderate and high transmission. Thus broad-spectrum antibiotic treatment ***should*** *be given* with antimalarial drugs to all children with suspected severe malaria in areas of moderate and high transmission until a bacterial infection is excluded. After the start of antimalarial treatment, unexplained deterioration may result from a supervening bacterial infection. Enteric bacteria (notably *Salmonella*) predominated in many trial series in Africa, but a variety of bacteria have been cultured from the blood of patients with a diagnosis of severe malaria.

Patients with secondary pneumonia or with clear evidence of aspiration should be given empirical treatment with an appropriate broad-spectrum antibiotic. In children with persistent fever despite parasite clearance, other possible causes of fever should be excluded, such as systemic *Salmonella* infections and urinary tract infections, especially in catheterized patients. In the majority of cases of persistent fever, however, no other pathogen is identified after parasite clearance. Antibiotic treatment should be based on culture and sensitivity results or, if not available, local antibiotic sensitivity patterns.

7.10.5 | USE OF ANTICONVULSANTS

The treatment of convulsions in cerebral malaria with intravenous (or, if this is not possible, rectal) benzodiazepines or intramuscular paraldehyde is similar to that for repeated seizures from any cause. In a large, double-blind, placebo-controlled evaluation of a single prophylactic intramuscular injection of 20 mg/kg bw of phenobarbital to children with cerebral malaria, the frequency of seizures was reduced but the mortality rate was increased significantly. This resulted from respiratory arrest and was associated with additional use of benzodiazepine. *A 20 mg/kg bw dose of phenobarbital should **not** be given without respiratory support.* It is not known whether a lower dose would be effective and safer or whether mortality would not increase if ventilation were given. In the absence of further information, prophylactic anticonvulsants are not recommended.

7.10.6 | TREATMENTS THAT ARE NOT RECOMMENDED

In an attempt to reduce the high mortality from severe malaria, various adjunctive treatments have been evaluated, but none has proved effective and many have been shown to be harmful. Heparin, prostacyclin, desferroxamine, pentoxifylline, low-molecular-mass dextran, urea, high-dose corticosteroids, aspirin anti-TNF antibody, cyclosporine A, dichloroacetate, adrenaline, hyperimmune serum, *N*-acetylcysteine and bolus administration of albumin are not recommended. In addition, use of corticosteroids increases the risk for gastrointestinal bleeding and seizures and has been associated with prolonged coma resolution times when compared with placebo.

7.11 | TREATMENT OF SEVERE MALARIA DURING PREGNANCY

Women in the second and third trimesters of pregnancy are more likely to have severe malaria than other adults, and, in low-transmission settings, this is often complicated by pulmonary oedema and hypoglycaemia. Maternal mortality is approximately 50%, which is higher than in non-pregnant adults. Fetal death and premature labour are common.

Parenteral antimalarial drugs should be given to pregnant women with severe malaria in full doses without delay. *Parenteral artesunate is the treatment of choice in all trimesters*. Treatment must not be delayed. If artesunate is unavailable, intramuscular artemether should be given, and if this is unavailable then parenteral quinine should be started immediately until artesunate is obtained.

Obstetric advice should be sought at an early stage, a paediatrician alerted and blood glucose checked frequently. Hypoglycaemia should be expected, and it is often recurrent if the patient is receiving quinine. Severe malaria may also present immediately after delivery. Postpartum bacterial infection is a common complication and should be managed appropriately.

7.12 | TREATMENT OF SEVERE *P. VIVAX* MALARIA

Although *P. vivax* malaria is considered to be benign, with a low case-fatality rate, it may cause a debilitating febrile illness with progressive anaemia and can also occasionally cause severe disease, as in *P. falciparum* malaria. Reported manifestations of severe *P. vivax* malaria include severe anaemia, thrombocytopenia, acute pulmonary oedema and, less commonly, cerebral malaria, pancytopenia, jaundice, splenic rupture, haemoglobinuria, acute renal failure and shock.

Prompt effective treatment and case management should be the same as for severe *P. falciparum* malaria (see section 7.4). Following parenteral artesunate, treatment can be completed with a full treatment course of oral ACT or chloroquine (in countries where chloroquine is the treatment of choice). A full course of radical treatment with primaquine should be given after recovery.

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8 | MANAGEMENT OF MALARIA CASES IN SPECIAL SITUATIONS

8.1 | EPIDEMICS AND HUMANITARIAN EMERGENCIES

Environmental, political and economic changes, population movement and war can all contribute to the emergence or re-emergence of malaria in areas where it was previously eliminated or well controlled. The displacement of large numbers of people with little or no immunity within malaria-endemic areas increases the risk for malaria epidemics among the displaced population, while displacement of people from an endemic area to an area where malaria has been eliminated can result in re-introduction of transmission and a risk for epidemics in the resident population.

Climate change may also alter transmission patterns and the malaria burden globally by producing conditions that favour vector breeding and there by increasing the risks for malaria transmission and epidemics.

8.1.1 | PARASITOLOGICAL DIAGNOSIS DURING EPIDEMICS

In the acute phase of epidemics and complex emergency situations, facilities for laboratory diagnosis with good-quality equipment and reagents and skilled technicians are often not available or are overwhelmed. Attempts should be made to improve diagnostic capacity rapidly, including provision of RDTs. If diagnostic testing is not feasible, the most practical approach is to treat all febrile patients as suspected malaria cases, with the inevitable consequences of over-treatment of malaria and potentially poor management of other febrile conditions. If this approach is used, it is imperative to monitor intermittently the prevalence of malaria as a true cause of fever and revise the policy appropriately. This approach has sometimes been termed “mass fever treatment”. This is not the same as and should not be confused with “mass drug administration”, which is administration of a complete treatment course of antimalarial medicines to every individual in a geographically defined area without testing for infection and regardless of the presence of symptoms (see section 10).

8.1.2 | MANAGEMENT OF UNCOMPLICATED FALCIPARUM MALARIA DURING EPIDEMICS

The principles of treatment of uncomplicated malaria are the same as those outlined in section 4. Active case detection should be undertaken to ensure that as many patients as possible receive adequate treatment, rather than relying on patients to come to a clinic.

8.1.3 | MANAGEMENT OF SEVERE MALARIA DURING EPIDEMICS

The principles of treatment of severe malaria during epidemics are the same as those outlined in section 7. In humanitarian emergencies, when there are many patients and many present late, effective triage, with immediate resuscitation and treatment, are essential. In epidemic situations, severe malaria is often managed in temporary clinics or in situations in which staff shortages and the high workload make intensive case monitoring difficult.

8.1.4 | EPIDEMICS OF MIXED FALCIPARUM AND VIVAX OR VIVAX MALARIA

ACTs (except artesunate + SP) should be used to treat uncomplicated malaria in mixed-infection epidemics, as they are highly effective against all malaria species. In areas with pure *P. vivax* epidemics, ACTs or chloroquine (if prevalent strains are sensitive) should be used.

8.1.5 | ANTI-RELAPSE THERAPY FOR *P. VIVAX* MALARIA

Administration of 14-day primaquine anti-relapse therapy for vivax malaria may be impractical in epidemic situations because of the duration of treatment and the difficulty of ensuring adherence. If adequate records are kept, therapy can be given in the post-epidemic period to patients who have been treated with blood schizontocides.

8.2 | MALARIA ELIMINATION SETTINGS

8.2.1 | USE OF GAMETOCYTOCIDAL DRUGS TO REDUCE TRANSMISSION

ACT reduces *P. falciparum* gametocyte carriage and transmission markedly, but this effect is incomplete, and patients presenting with gametocytaemia may be infectious for days or occasionally weeks, despite ACT. The strategy of using a single dose of primaquine to reduce infectivity and thus *P. falciparum* transmission has been widely used in low transmission settings.

Use of primaquine as a *P. falciparum* gametocytocide has a particular role in programmes to eliminate *P. falciparum* malaria. The population benefits of reducing malaria transmission by gametocytocidal drugs require that a high proportion of patients receive these medicines. WHO recommends the addition of a single dose of primaquine (0.25

mg base/kg bw) to ACT for uncomplicated falciparum malaria as a gametocytocidal medicine, particularly as a component of pre-elimination or elimination programmes. A recent review of the evidence on the safety and effectiveness of primaquine as a gametocytocide of *P. falciparum* indicates that a single dose of 0.25 mg base/kg bw is effective in blocking infectivity to mosquitos and is unlikely to cause serious toxicity in people with any of the G6PD variants. Thus, the G6PD status of the patient does not have to be known before primaquine is used for this indication.

8.3 | ARTEMISININ-RESISTANT FALCIPARUM MALARIA

Artemisinin resistance in *P. falciparum* is now prevalent in parts of Cambodia, the Lao People's Democratic Republic, Myanmar, Thailand and Viet Nam. There is currently no evidence for artemisinin resistance outside these areas. The particular advantage of artemisinins over other antimalarial drugs is that they kill circulating ring-stage parasites and thus accelerate therapeutic responses. This is lost in resistance to artemisinin. As a consequence, parasite clearance is slowed, and ACT failure rates and gametocytaemia both increase. The reduced efficacy of artemisinin places greater selective pressure on the partner drugs, to which resistance is also increasing. This situation poses a grave threat. In the past chloroquine resistant parasites emerged near the Cambodia–Thailand border and then spread throughout Asia and Africa at a cost of millions of lives. In Cambodia, where artemisinin resistance is worst, none of the currently recommended treatment regimens provides acceptable cure rates (> 90%), and continued use of ineffective drug regimens fuels the spread of resistance. In Cambodia use of atovaquone–proguanil instead of ACT resulted in very rapid emergence of resistance to atovaquone.

In this dangerous, rapidly changing situation, local treatment guidelines cannot be based on a solid evidence base; however, the risks associated with continued use of ineffective regimens are likely to exceed the risks of new, untried regimens with generally safe antimalarial drugs. At the current levels of resistance, the artemisinin derivatives still provide significant antimalarial activity; therefore, longer courses of treatment with existing or new augmented combinations or treatment with new partner medicines (e.g. artesunate + pyronaridine) may be effective. Studies to determine the best treatments for artemisinin-resistant malaria are needed urgently.

It is strongly recommended that single-dose primaquine (as a gametocytocide) be added to all falciparum malaria treatment regimens (section 4.5). For the treatment of severe malaria in areas with established artemisinin resistance, it is recommended that parenteral artesunate and parenteral quinine be given together in full doses, as described in section 7.4.

3RD

EDITION

9 | ARTEMISININ-BASED COMBINATION THERAPIES NOT CURRENTLY RECOMMENDED FOR GENERAL USE

The pipeline for new antimalarial drugs is healthier than ever before, and several new compounds are in various stages of development. Some novel antimalarial agents are already registered in some countries. The decision to recommend antimalarial drugs for general use depends on the strength of the evidence for safety and efficacy and the context of use. In general, when there are no satisfactory alternatives, newly registered drugs may be recommended; however, for global or unrestricted recommendations, considerably more evidence than that submitted for registration is usually required, to provide sufficient confidence for their safety, efficacy and relative merits as compared with currently recommended treatments.

Several new antimalarial drugs or new combinations have been introduced recently. Some are still in the pre-registration phase and are not discussed here. Artesunate + pyronaridine, arterolane + piperaquine, artemisinin + piperaquine base and artemisinin + napththoquine are new ACTs, which are registered and used in some countries. In addition, there are several new generic formulations of existing drugs. None of these yet has a sufficient evidence base for general recommendation (i.e. unrestricted use).

A systematic review of **artesunate + pyronaridine** included six trials with a total of 3718 patients. Artesunate + pyronaridine showed good efficacy as compared with artemether + lumefantrine and artesunate + mefloquine in adults and older children with *P. falciparum* malaria, but the current evidence for young children is insufficient to be confident that the drug is as effective as currently recommended options. In addition, regulatory authorities noted slightly higher hepatic transaminase concentrations in artesunate + pyronaridine recipients than in comparison groups and recommended further studies to characterize the risk for hepatotoxicity. Preliminary data from repeat-dosing studies are reassuring (see Annex 4, A4.17).

Arterolane + piperaquine is a combination of a synthetic ozonide and piperaquine phosphate that is registered in India for use only in adults There are currently insufficient data to make general recommendations.

Artemisinin + piperaquine base combines two well-established, well-tolerated compounds. It differs from previous treatments in that the piperaquine is in the base form, the artemisinin dose is relatively low, and the current labelling is for only a 2-day regimen. There are insufficient data from clinical trials for a general recommendation, and there is concern that the artemisinin dose regimen provides insufficient protection against resistance to the piperaquine component.

Artemisinin + naphthoquine is also a combination of two relatively old compounds that is currently being promoted as a single-dose regimen, contrary to WHO advice for 3 days of the artemisinin derivative. There are currently insufficient data from rigorously conducted randomized controlled trials to make general recommendations (see Annex 4, A4.18).

Many ACTs are generics. The bioavailability of generics of currently recommended drugs must be comparable to that of the established, originally registered product, and the satisfactory pharmaceutical quality of the product must be maintained.

3RD

EDITION

10 | MASS DRUG ADMINISTRATION

Mass antimalarial drug administration has been used extensively in various forms over the past 80 years. The objective is to provide therapeutic concentrations of antimalarial drugs to as large a proportion of the population as possible in order to cure any asymptomatic infections and also to prevent reinfection during the period of post-treatment prophylaxis.[9] Mass drug administration rapidly reduces the prevalence and incidence of malaria in the short term, but more studies are required to assess its longer-term impact, the barriers to community uptake, and its potential contribution to the development of drug resistance.[10] In an analysis of 38 mass drug administration projects carried out since 1932,[11] only one was reported to have succeeded in interrupting malaria transmission permanently. In this study, chloroquine, SP and primaquine were provided weekly to the small population of Aneityum Island in Vanuatu for 9 weeks before the rainy season, in combination with distribution of insecticide-treated nets.[12]

There is considerable divergence of opinion about the benefits and risks of mass antimalarial drug administration. As a consequence, it has been little used in recent years; however, renewed interest in malaria elimination and the emerging threat of artemisinin resistance has been accompanied by reconsideration of mass drug administration as a means for rapidly eliminating malaria in a specific region or area. During mass campaigns, every individual in a defined population or geographical area is requested to take antimalarial treatment at approximately the same time and at repeated intervals in a coordinated manner. This requires extensive community engagement to achieve a high level of community acceptance and participation. The optimum timing depends of the elimination kinetics of the antimalarial (e.g. in current initiatives using dihydroartemisinin + piperaquine, the drug is given monthly for 3 months at treatment doses, as the residual piperaquine levels suppress reinfections for 1 month). Depending on the contraindications for the medicines used, pregnant women, young infants and other population groups may be excluded from the campaign. Thus, the drugs used, the number of treatment rounds, the optimum intervals and the support structures necessary are all context-specific and are still subject to active research. In the past, vivax elimination programmes were based on pre-seasonal mass radical treatment with primaquine (0.25 mg/kg/for 14 days) without testing for G6PD deficiency or monitoring primaquine-induced haemolysis, although in some cases interrupted regimens were used: 4 days' treatment, 3 days of no treatment, then continuation to complete the course (usually 11 days) if the drug was well tolerated.[13]

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[9] White NJ. How antimalarial drug resistance affects post-treatment prophylaxis. Malar J 2008;11:7–9.

[10] Poirot E, Skarbinski J, Sinclair D, Kachur SP, Slutsker L, Hwang J. Mass drug administration for malaria. Cochrane Database Systemat Rev 2013;11:CD008846. doi: 10.1002/14651858.CD008846.pub2.

[11] von Seidlein L, Greenwood BM. Mass administration of antimalarial drugs. Trends Parasitol 2003;19:790–6.

[12] Kaneko A, Taleo G, Kalkoa M, Yamar S, Kobayakawa T, Björkman A. Malaria eradication on islands. Lancet 2000;356:1560–4.

[13] Kondrashin A, Baranova AM, Ashley EA, Recht J, White NJ, Sergiev VP. Mass primaquine treatment to eliminate vivax malaria: lessons from the past. Malar J 2014;13:51.

Both mass drug administration and mass screening and treatment require that a high proportion of the target population is reached by the intervention. This requires informed, enthusiastic community participation and comprehensive support structures.

Once mass drug administration is terminated, if malaria transmission is not interrupted or importation of malaria is not prevented, then malaria endemicity in the area will eventually return to its original levels (unless the vectorial capacity is reduced in parallel and maintained at a very low level). The time it takes to return to the original levels of transmission will depend on the prevailing vectorial capacity.[14] If malaria is not eliminated from the target population, then mass drug administration may provide a significant selective pressure for the emergence of resistance. The rebound in malaria may be associated temporarily with higher morbidity and mortality if drug administration was maintained long enough for people to lose herd immunity against malaria. For this reason, mass drug administration should not be started unless there is a good chance that focal elimination will be achieved. In some circumstances (e.g. containment of artemisinin-resistant *P. falciparum*), elimination of only one species may be the objective.

[14] Vectorial capacity: number of new infections that the population of a given anopheline mosquito vector would distribute per case per day at a given place and time, assuming the human population is not immune. Factors affecting vectorial capacity include: the density of female anophelines relative to humans; their longevity, frequency of feeding and propensity to bite humans; and the length of the extrinsic (i.e. in the mosquito) cycle of the parasite.